

The Crypto Cycle and Institutional Investors^{*}

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ABSTRACT

We examine aggregate fluctuations in crypto markets and their relationship to global equity markets and US monetary policy. First, we document that changes in the correlation between crypto and global equity markets can be explained by changes in the participation of institutional investors in crypto markets. Second, we find that US monetary policy significantly affects crypto markets, but only when the participation of institutional investors is high. Third, within high-participation periods, the response is concentrated where the leverage of crypto investors, measured by the perpetual-futures funding rate, is high. Finally, we rationalize our empirical results in a heterogeneous-agent model in which large investors holding both asset classes create a direct link between them, and the effective risk aversion of the crypto market is governed by their balance-sheet constraints.

Keywords: Institutional Investors; Crypto Assets; US Monetary Policy.

JEL classification: G23; E44; E52; F33.

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1 Introduction

Crypto assets vary substantially in their design and value propositions, yet their prices have moved in common cycles.¹ Total crypto market capitalization boomed from US\$20 billion in 2016 to almost US\$3 trillion in November 2021, then collapsed to below US\$1 trillion in the latest crypto “winter” before rebounding to over US\$4 trillion.² The extent to which crypto markets move in tandem with other asset classes has also fluctuated. While crypto returns have generally correlated with broad market returns (Liu and Tsyvinski, 2021), Bitcoin initially provided a partial hedge against market risk, before becoming increasingly correlated with the S&P 500 from 2020 (Adrian, Iyer, and Qureshi, 2022). This correlation in turn subsequently weakened during the latest crypto “winter”, as we show below.

In parallel to these cycles, the global crypto investor base has changed substantially. From an initial user base of technology enthusiasts, periods of exponential returns have attracted both a wider range of retail investors and new, generally much larger, institutional investors.³ Subsequent crashes have led to increasing pushback from politicians, scrutiny from regulators, and waning interest from some investors.⁴ These fluctuations in the composition of the crypto investor base could have important implications for crypto prices. For instance, as more institutional investors enter crypto markets, the marginal investors in crypto and equities become increasingly similar, potentially increasing the co-movement between the two asset classes.

In this paper, we examine how the composition of the crypto investor base—specifically, the participation of institutional investors—determines the transmission of global risk factors and US monetary policy to crypto prices. We document three main results. First, the correlation between

¹For instance, the white papers of prominent crypto assets include aims to provide peer-to-peer electronic cash, more efficient transactions, censorship-resistant decentralized computing, and functionality within a financial services ecosystem (Nakamoto, 2008; Buterin, 2014; Ripple Labs Inc., 2014; Binance, 2017). We exclude stablecoins from our analysis, as they are intended to maintain a constant price.

²Source: coinmarketcap.com.

³See Benetton and Compiani (2024); Auer and Tercero-Lucas (2022); Auer, Farag, Lewrick, Orazem, and Zoss (2022); Aiello, Baker, Balyuk, Di Maggio, Johnson, and Kotter (2023).

⁴See, for example, “Crypto.com Winding Down U.S. Institutional Business.” by Helene Braun on Coindesk, August 14, 2023. [Link](#).

crypto and global equity markets co-moves with the participation of institutional investors in crypto markets. Lagged institutional trading shares explain a substantial part of this correlation. Consistent with this, the correlation between Bitcoin and equities rose, relative to that for Ether, after the SEC approved spot Bitcoin ETFs in January 2024 and thereby increased institutional investors' access to Bitcoin. Second, US monetary policy significantly affects the crypto cycle, but only when institutional participation is high. Third, within high-participation periods, the response is concentrated where the leverage of crypto investors is high, as measured by the perpetual-futures funding rate. These results admit a simple interpretation. Large investors holding both asset classes create a direct link between them. Because these investors are levered, a monetary tightening erodes their risk-bearing capacity and drags crypto prices down together with equities. We rationalize this interpretation in a heterogeneous-agent model in which the effective risk aversion of the crypto market is governed by the balance-sheet constraints of levered institutional investors.

We begin our analysis by providing stylized facts on aggregate movements in crypto markets. Using a dynamic factor model on daily data from January 2018 to March 2026, we identify a single common component in the prices of the major crypto assets—the “crypto factor”—that explains 67% of the variance in crypto prices, which is substantially larger than the 20% observed for global equities by Miranda-Agrippino and Rey (2020).⁵ This difference highlights the greater concentration of market capitalization in the largest crypto assets relative to that in the largest equities. We then compare the crypto factor to a set of global equity factors (in the spirit of Rey, 2013; Miranda-Agrippino and Rey, 2020) and find a positive correlation over almost the entire sample, driven by a particularly strong correlation between 2020 and the beginning of 2023. Importantly, the increasing co-movement over that period is not limited to Bitcoin vis-a-vis the S&P 500 but pertains more broadly to the crypto and global equity factors. We also find that the crypto factor correlates most strongly with the global tech factor and the small-cap factor, while it is surprisingly less correlated with the global financial factor.

⁵Throughout the paper, the *crypto factor* refers to this estimated common component of crypto prices, and the *crypto cycle* refers to its fluctuations over time—analogueous to the global financial cycle of Rey (2013).

This increased correlation between crypto and equities coincided with substantial growth in the participation of institutional investors. To formally investigate this relationship, we first regress daily changes in the crypto factor on changes in the equity factor interacted with lagged values of the trading share of institutional investors—i.e., we test whether the share of institutional trading explains the one-day-ahead correlation between crypto and equity markets. We find that a one-standard-deviation increase in the lagged institutional trading share corresponds to a co-movement between crypto and global equity markets that is roughly twice as large. This finding is robust across various proxies for institutional participation, including measures based on on-chain transaction size and on off-chain trading activity reported by Coinbase, one of the most important centralized exchanges.

We confirm this relationship using an alternative, complementary methodology that exploits the Securities and Exchange Commission’s (SEC) approval of Bitcoin Exchange-Traded Funds (ETFs) in January 2024. This lowered the cost to institutions of gaining exposure to Bitcoin, relative to the cost of gaining exposure to other crypto assets. We use a difference-in-differences approach to compare the change in the Bitcoin-equities correlation before vs. after the SEC approval to the equivalent change in the Ether-equities correlation. Using different specifications and time windows, we find robust evidence that the correlation between Bitcoin and equities increased, relative to the correlation between Ether and equities, in line with our hypothesis that institutional entry increases co-movement between crypto and equity markets.

The role of institutional investors in the correlation between the crypto and global equity cycle raises the question of whether they also influence crypto’s response to monetary policy. Since US monetary policy affects the global financial cycle (Miranda-Agrippino and Rey, 2020), we first explore whether a similar relationship holds for the crypto factor. We identify US monetary-policy shocks using the high-frequency surprises of Acosta, Ajello, Bauer, Loria, and Miranda-Agrippino (2025), measured in narrow windows around FOMC announcements, and estimate local projections on announcement days. Because the surprises are measured from interest-rate moves in a narrow

window around each announcement, they are identified independently of broader daily asset-price movements, addressing the simultaneity concerns inherent in daily financial data. We find that US monetary policy affects the crypto cycle, contrasting starkly with claims that crypto assets provide a hedge against market risk: the crypto factor falls by 0.41 standard deviations per one-percentage-point surprise on impact, and the response remains negative in the days that follow. These results also hold in a daily VAR with the surprise as an internal instrument, and in an event study around FOMC announcements. Notably, as with the global equity cycle, the transmission appears specific to US monetary policy: comparable high-frequency ECB and Bank of England surprises generate no negative response of the crypto factor, likely reflecting that crypto markets are highly dollarized.⁶

Having established that US monetary policy affects the crypto cycle, we examine the role of institutional investors in its transmission. Specifically, we estimate regime-interacted local projections in which the response varies smoothly with the lagged institutional-investor share, controlling for a post-2020 dummy and its interaction with the shock so that the regime split is identified over and above any generic post-2020 shift in the monetary-policy response. We find that US monetary policy has a significant impact on crypto markets only when the institutional share is high: on impact, the high-regime response is -1.04 standard deviations per percentage point, while the low-regime response is close to zero, and the difference between the two is statistically significant. To provide direct evidence on the mechanism, we then measure the leverage of crypto investors at high frequency using the Bitcoin perpetual-futures funding rate—the shadow cost of levered positioning—and interact the two states: the negative response is concentrated where high institutional participation coincides with high leverage. This ties the transmission of US monetary policy to the risk-bearing capacity of levered institutional investors, and corroborates their importance for synchronization between global equity and crypto markets, since US monetary policy is an important driver of fluctuations in global equity markets (Miranda-Agrippino and Rey, 2020).

Finally, we formalize our empirical results using a heterogeneous-agent framework in which the

⁶For instance, the two largest stablecoins Tether and USD Coin are pegged to the dollar, while Coinbase, one of the largest centralized crypto exchanges, is listed on the Nasdaq.

participation of institutional investors in crypto markets affects the expected returns from crypto relative to equities. We assume two types of investors: retail investors who only invest in crypto assets, and institutional investors who can invest in both stocks and crypto assets. Retail investors are risk-averse, while institutional investors are risk-neutral but face a value-at-risk constraint. The equilibrium returns on equity assets are proportional to their variance and their covariance with crypto returns, and are inversely related to the institutional investors' risk-taking capacity (i.e., value-at-risk constraint). Similarly, we can rewrite the equilibrium returns on the crypto assets as a linear combination of their variance and their covariance with stocks' returns, scaled by the aggregate effective risk aversion, a wealth-weighted average of the risk aversion of retail crypto investors and the shadow cost of the institutional investors' value-at-risk constraint. This means that the higher the relative wealth of institutional investors, the higher the synchronization between the expected crypto and equity returns. Furthermore, since the presence of institutional investors in crypto markets increases the risk capacity of the marginal crypto investor, we interpret the higher reaction of crypto prices to monetary contraction as consistent with the fact that more levered investors are more sensitive to the economic cycle (see also Coimbra, Kim, and Rey, 2022; Adrian and Shin, 2014).

In conclusion, we offer a parsimonious explanation for several empirical results concerning the relationships between crypto prices, equity prices, and monetary policy. We show that the varying correlation between crypto and equity markets can be explained by changes in the composition of the crypto investor base. Furthermore, US monetary policy tightening significantly affects crypto markets, as for equities, when there is substantial institutional participation in crypto markets. By highlighting the role of institutional investors in crypto markets, our results suggest that policy-makers should monitor their participation closely. While it is true that crypto markets could help investors diversify their portfolios, higher institutional participation could also increase spillover risks between traditional and crypto markets.

The rest of this paper proceeds as follows. Section 2 relates our contributions to the literature.

Section 3 provides stylized evidence about the crypto and equity cycles, and tests the importance of institutional investors in their synchronization. Section 4 investigates the impact of US monetary policy on the crypto cycle and explores its relationship to institutional participation. Section 5 formalizes our empirical findings in a heterogeneous-agent model. Finally, Section 6 concludes.

2 Literature Review

We contribute to three main strands of literature. First, we build on the empirical literature studying the composition of crypto investors.⁷ Baur and Dimpfl (2018) argue the asymmetric volatility of crypto asset returns after positive vs. negative shocks can be explained by changes in the composition of the crypto investor base. Auer, Farag, Lewrick, Orazem, and Zoss (2022) are the first to focus on the role of institutional investors in crypto markets, and show that traditional financial institutions are starting to hold crypto assets. In particular, Auer et al. (2022) document that banks' exposure to crypto assets is low relative to their balance sheet, but it is significantly large when compared to trading volumes in crypto markets. Furthermore, non-bank financial institutions including hedge funds and asset managers are becoming an increasingly important source of revenue for crypto exchanges. Cornelli, Doerr, Frost, and Gambacorta (2023) also show differences in trading behavior between small and large investors during crisis episodes, highlighting their potentially important role in determining prices. We contribute to this literature by studying how institutional participation in crypto markets influences the synchronization of crypto markets with traditional financial markets.

Second, our work builds on a strand of literature that assesses the impact of macroeconomic news events, including monetary policy announcements, on Bitcoin returns (Corbet, Larkin, Lucey,

⁷For instance, Auer and Tercero-Lucas (2022) study the profile of US crypto investors and highlight that they are in general less motivated by distrust in the traditional financial system than by the prospects for high returns. Similarly, Hackethal, Hanspal, Lammer, and Rink (2022) and Didisheim, Frascini, and Somoza (2022) document the behavior of crypto retail investors and their portfolio allocation between equity and crypto assets. Using transaction-level data from millions of US households, Aiello et al. (2023) show that crypto investors, except for high-income early adopters, resemble the general population, spanning all income levels with significant contributions from high-income individuals. Relatedly, Campello, Gallo, Mota, and Terracciano (2026) document that even investors' demand for safety is largely met within the crypto ecosystem by stablecoin deposits in DeFi.

Meegan, and Yarovaya, 2020; Pyo and Lee, 2020; Ma, Tian, Hsiao, and Deng, 2022; Elsayed and Sousa, 2024; Karau, 2023). Benigno and Rosa (2023) use intraday data to investigate whether macroeconomic news affects the Bitcoin price and conclude that they are uncorrelated; we differ in considering potential impacts over a longer horizon and by investigating the potential role of institutional investors. Our paper considers the underlying drivers of co-movement in the entire asset class and the changing relationship with the global financial cycle. In doing so, we also build on Iyer (2022), who provides evidence of the positive correlation between equity markets and Bitcoin and Ether prices, and Liu and Tsyvinski (2021) who do similarly for an aggregate of all crypto assets.⁸ We contribute by documenting the extent to which monetary policies affect the aggregate crypto cycle and by uncovering the role of institutional investors and their balance-sheet constraints in shaping that transmission.

Third, our paper builds on the literature on the global financial cycle pioneered by the seminal contribution of Rey (2013). In more recent works, Miranda-Agrippino and Rey (2020) and Miranda-Agrippino and Rey (2022) highlight how US monetary policy affects this global financial cycle: a change in interest rates forces financial intermediaries to change their leverage and thus changes the effective risk appetite of the marginal investor; a US monetary contraction thus reduces global equity prices. We find that the same dynamics apply to crypto markets, contrasting starkly with views that they are uncorrelated with the traditional economy and an effective hedge against market risk. Indeed, we argue that crypto assets have become increasingly integrated with the global financial cycle. Within this literature, we also show that a standard asset pricing model with heterogeneous agents can rationalize our results. Specifically, we draw from the models in which the differences in risk aversion across agents determines the aggregate risk of the economy (Coimbra and Rey, 2024; Coimbra et al., 2022), and the cyclicity of leverage depends on the constraints of financial intermediaries (Adrian and Shin, 2014).⁹ Our funding-rate results provide direct evidence for this constraint channel in crypto markets.

⁸These papers in turn pick up from earlier work on co-movement and financial integration across equity markets in different countries (e.g., Forbes and Rigobon, 2002).

⁹See also Kekre and Lenel (2022) and Gourinchas, Rey, and Govillot (2010).

3 Crypto-Equity Synchronization

In this section, we first present a set of stylized facts about the crypto markets and how they can be exhaustively described by a single price component—i.e., the crypto factor—which reflects the crypto cycle. Next, we study the relation between the crypto and global equity cycles and the role of institutional investors in their synchronization.

3.1 Stylized Facts about the Crypto Cycle

The prices of crypto assets are highly correlated. Among the crypto assets with the largest market capitalization, the pairwise correlations of daily returns are remarkably high: Bitcoin, for instance, has an average correlation of 60% with the other major assets.¹⁰ These correlations are also much larger than those documented across equity markets (see, for instance, Rey, 2013). We thus conjecture the existence of a common crypto factor that co-moves with crypto prices, in the same spirit as the global equity factor pioneered by Rey (2013).

To summarize the fluctuations in crypto markets in one variable, we use dynamic factor modeling, a dimensionality reduction technique.¹¹ We label the resulting common factor as the *crypto factor* which synthesizes the crypto cycle and therefore the fluctuations in crypto markets. Figure 1a shows the crypto factor and the underlying price series from which we extract it. The crypto factor effectively captures the salient phases that characterized the crypto cycle—such as the slumps after the collapses of Terra and FTX in 2022 and the booms after the Bitcoin ETF approval and the re-election of President Trump—without being overly influenced by isolated spikes like those of Ripple and Tron.

To gauge the importance of this factor more systematically, we regress each price series in turn on the crypto factor. On average, 67% of variation in the underlying series is explained by our crypto factor.¹² The R^2 is 91% for Bitcoin and lies between 47% and 98% for all assets other than

¹⁰We show the full correlation matrix for the largest crypto assets in Figure B.1 in Appendix B.

¹¹Refer to Appendix A for a detailed explanation of the factor construction.

¹²See Appendix A Figure A.1 for the breakdown across individual crypto assets.

Cardano, underscoring the high degree of co-movement over our sample period. For comparison, the global equity factor calculated by Miranda-Agrippino and Rey (2020) explains only 20% of global equity prices, highlighting the greater co-movement and concentration of market capitalization in the crypto market. These results strongly suggest that a single crypto factor is sufficient to explain the crypto cycle, i.e., the aggregate fluctuations of crypto markets.

3.2 The Crypto and Global Equity Cycles

We now turn to the relationship between the crypto factor and global equities. Iyer (2022) documents an increase in correlation between Bitcoin and S&P 500 returns from 2020 to 2022. We therefore conjecture that crypto markets may have become more integrated and synchronized with the equity cycle. To assess this, in this subsection we compute a global equity factor then examine its relationship to the crypto factor.

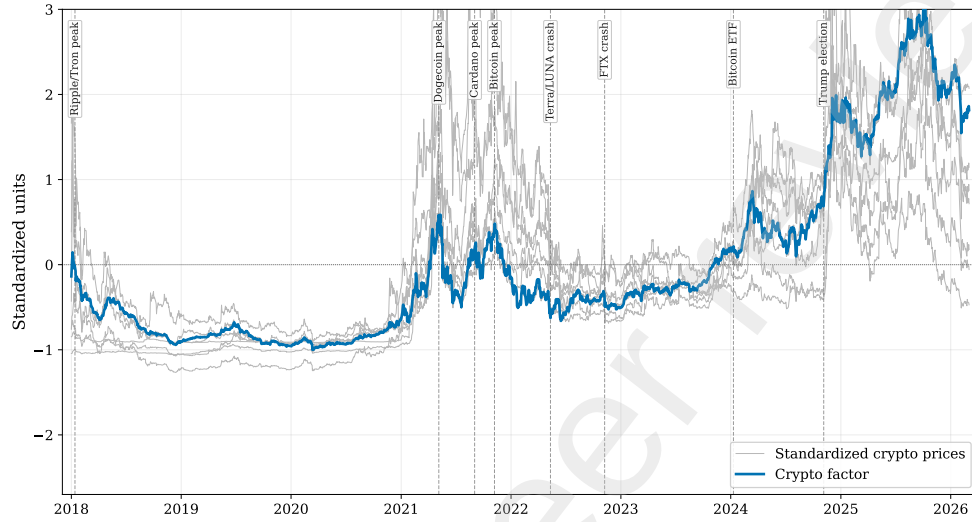
We construct the global equity factor using all the equity indices available on Eikon-Thomson Reuters for the largest fifty countries by GDP.¹³ We follow Miranda-Agrippino and Rey (2020) and the same methodology as in the previous section to compute: a general factor using all major stock indices, a factor for small capitalization stocks, and separate factors for each of the technology and financial sectors. Figure 1b shows both the equity and the crypto factors. As with the crypto factor, the equity factor reliably replicates the dynamics of global markets, with the sharp decline during the COVID-19 shock, the subsequent recovery and the downturn in early 2022. Generally speaking, the two series are fairly uncorrelated at the beginning of the sample and become increasingly correlated thereafter, with the strength of the co-movement fluctuating over time.

In Table 1, we regress changes in the crypto factor on changes in each of the other factors. Column (1) shows that, in general, the correlation between the crypto and the equity factor is highly significant, while columns (2) and (7) specifically highlight that this relationship is driven by the technology and small-cap components.

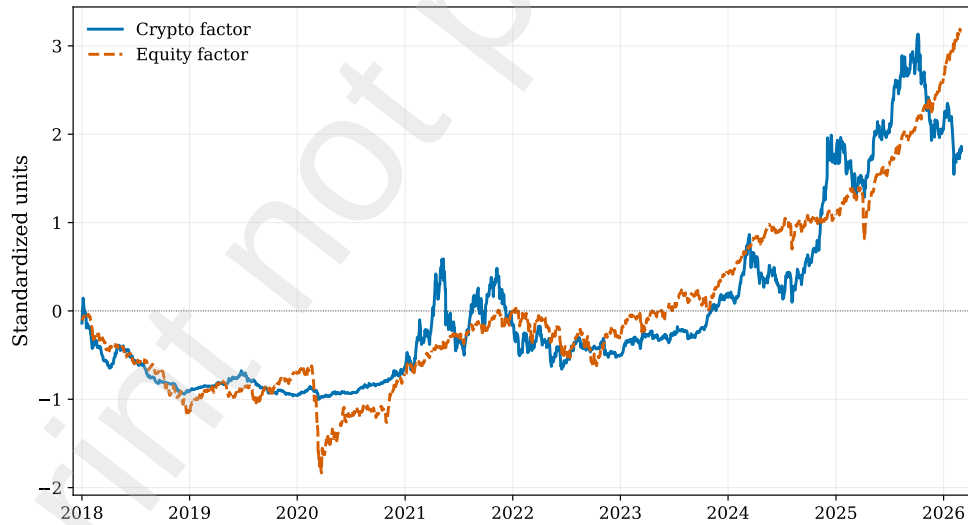
¹³Table A.1 in Appendix A details the full list of indices used.

Figure 1: The Crypto and Global Equity Cycles

(a) The crypto factor and the underlying crypto prices



(b) The crypto and global equity factors



Notes: Panel (a) shows the crypto factor (blue) and the standardized crypto prices from which it is constructed (gray) using a dynamic factor model; vertical dashed lines mark salient market events. Panel (b) shows the standardized time series of the crypto and equity factors, derived using dynamic factor modeling from a large range of crypto prices and equity indices respectively, as described in Section 3.1. Daily data, January 2018–March 2026; crypto prices from CryptoCompare, equity indices from Eikon/Thomson Reuters.

Table 1: Factor Regressions

	Δ Crypto Factor _t						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Δ Global Equity Factor (General) _t	0.216*** (7.48)						
Δ Global Tech Factor _t		0.324*** (10.09)				0.330*** (9.10)	
Δ Global Equity Factor excl. Tech _t			-0.003 (-0.08)				
Δ Global Financial Factor _t				0.177*** (5.82)		-0.011 (-0.33)	
Δ Global Equity Factor excl. Financials _t					0.126*** (4.28)		
Δ Global Small Caps Factor _t							0.218*** (7.98)
Constant	YES	YES	YES	YES	YES	YES	YES
Obs.	1993	1993	1993	1993	1993	1993	1993
R ²	0.045	0.102	0.000	0.030	0.016	0.102	0.046

Notes: Variables are standardized; *t*-statistics in parentheses. Daily data, January 2018–March 2026; crypto and equity factors constructed as described in Section 3, with equity indices from Eikon/Thomson Reuters. *, **, *** denote significance at the 10%, 5%, and 1% levels.

The evidence from simple correlations points in the same direction.¹⁴ Consistent with Iyer (2022), the correlation between Bitcoin and the S&P 500 was lower before 2020 but increased significantly afterward. The same holds for the correlation between the crypto and global equity factors.

3.3 Institutional Investors and Crypto-Equity Synchronization

Institutional investors have substantially increased their exposure to crypto assets over time. Using a supervisory database, Auer et al. (2022) document the growing importance of traditional financial institutions in crypto markets. They show that institutional exposure to crypto assets has increased, and that, while it remains small relative to their balance sheets, it is significant for the crypto market, which was previously populated predominantly by retail investors.

In this subsection, we explore the extent to which this dramatic rise and subsequent fluctuations in institutional investors' participation in crypto markets can explain the changing synchroniza-

¹⁴The full correlation matrices for a range of crypto, equity and macro-financial variables, before and after 2020, are reported in Figure B.2 in Appendix B, together with the *p*-values of the changes in correlations (Figure B.3), computed by regressing the different crypto factors on equity factors (or other variables, e.g., gold and oil prices), a time dummy and their interactions.

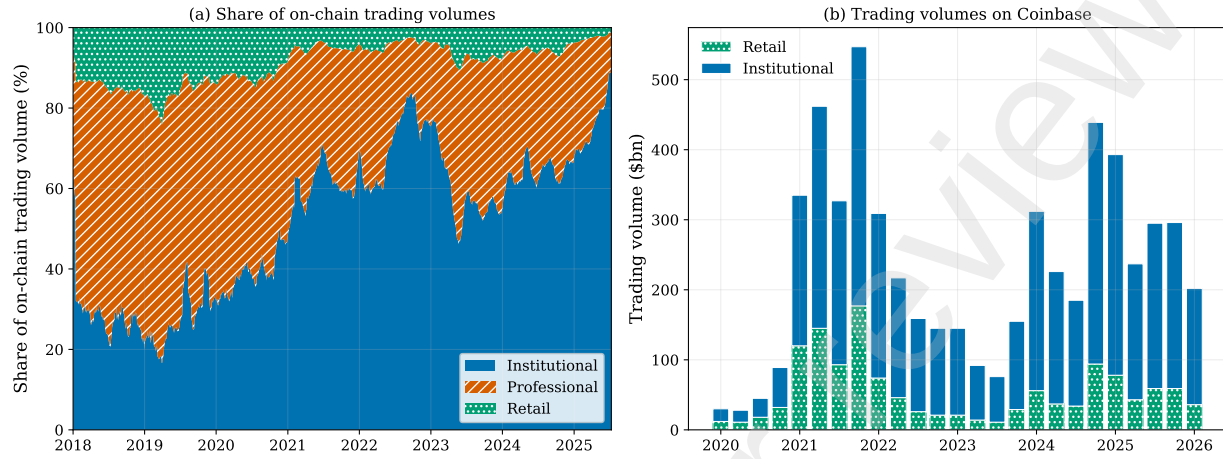
tion of crypto and equity markets. Given the opaque nature of crypto markets, where flows are either pseudonymous or occur within centralized exchanges, we use two complementary proxies for institutional participation. Both sources, one based on on-chain data and one on off-chain data, show similar aggregate trends and also align with the patterns documented in Auer et al. (2022). First, we use data from public blockchains assembled by Chainalysis, who classify investor types by transaction size: transfers above \$1 million are attributed to institutional investors, transfers under \$10,000 to retail investors, and those in between to professional investors. On this measure, institutional investors' share of crypto trading volumes rose dramatically after 2019 and then declined toward the end of 2023, as shown in Figure 2 Panel (a).¹⁵ Second, we observe a similar evolution in data from Coinbase, a major centralized exchange, which reports retail vs. institutional volumes in its public financial statements (Figure 2 Panel (b)). Despite the data sources reflecting different parts of the market, they both confirm that institutional trading volumes dramatically increased at the end of 2021 and then declined in 2023.

In Figure 3, we plot the 90-day rolling correlations of the crypto and equity factor, Bitcoin and S&P 500, and of the share of institutional investors. At the beginning of the sample, the crypto and equity markets were essentially uncorrelated. Thereafter, on average, the correlations started increasing, with the S&P 500–Bitcoin correlation reaching 60% in 2022. Eventually, in 2023 the correlation steadily decreased to 20%, before partially recovering over 2024–2025. In parallel, the share of institutional investors was rather low and flat in the first part of the sample and then firmly increased until the end of 2022, going from roughly 25% to 80%. As with the crypto-equity correlations, the share of institutional investors dropped in 2023 by more than 20 percentage points. Together, these patterns suggest that institutional investors might play a role in the correlation between crypto and global equity cycles.

In Table 2, we formally investigate the relationship between the share of institutional investors and the correlation between equity and crypto factors. In the absence of exogenous variation in the share of trading by institutional investors, we advance toward identification by (i) using lagged

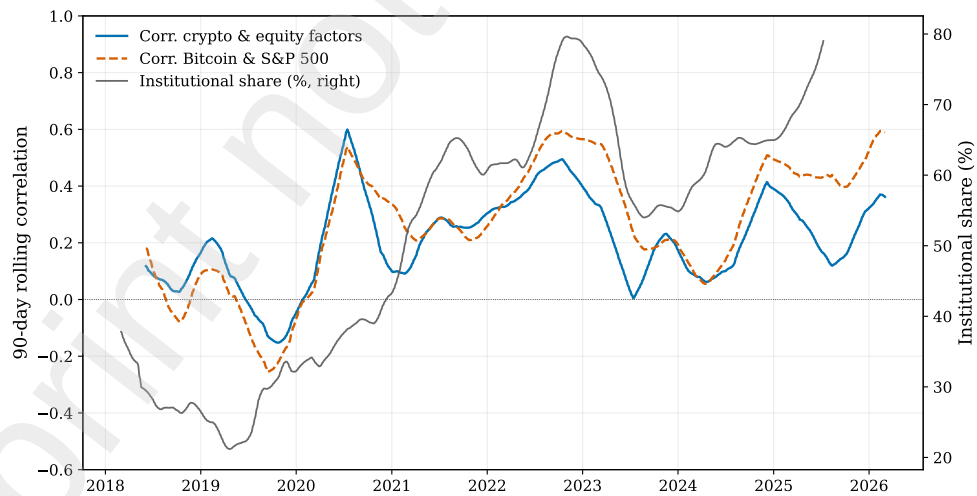
¹⁵ Absolute volumes also show a similar pattern.

Figure 2: Increasing Institutional Participation in Crypto Markets



Notes: Panel (a) shows the smoothed share of on-chain trading volumes by investor type over time, using data from public blockchains assembled by Chainalysis. Investor types are classified according to transaction sizes, i.e., trades under \$10k are classified as trades by retail investors, the ones from \$10k to \$1M by professional investors, and those above \$1M by institutional investors. Panel (b) shows the trading volumes on Coinbase of retail and institutional investors, reported in the company's public financial statements (available at <https://investor.coinbase.com/financials/quarterly-results/>). Daily (smoothed) data, 2018–July 2025, in Panel (a); quarterly data from Coinbase's public filings in Panel (b).

Figure 3: Rolling Correlations between Crypto and Equities



Notes: This graph shows the smoothed 90-day rolling correlations between the crypto and equity factors, the Bitcoin and the S&P 500, and the (smoothed) share of institutional investors trading crypto assets. Daily data, 2018–March 2026 (institutional share through July 2025); correlations smoothed by a 90-day moving average; crypto prices from CryptoCompare, equity indices from Eikon/Thomson Reuters, institutional share from Chainalysis.

values of the trading shares and (ii) controlling for year-week fixed effects. The former alleviates concerns of reverse causation—i.e., that institutional participation might be partially driven by the correlation between crypto and equity prices. The latter alleviates the concern that time-varying omitted factors drive both the institutional share and the crypto-equity correlation.

Table 2 column (1) shows that, over the full sample, there is a significant correlation between the crypto and equity factors. In column (2), we test the role of institutional investors by looking at the interactions with the lagged share of institutional investors (Figure 2 panel (a)).¹⁶ Our results show that a higher share of institutional investors trading crypto leads to a significantly higher correlation between equity and crypto markets. The magnitude is economically meaningful: a one-standard-deviation increase in the lagged institutional share roughly doubles the co-movement between the crypto and equity factors.

Our findings are robust to multiple specifications. In column (3), we compute the institutional share from exchange-related transactions only (centralized and decentralized exchanges), and in column (4) we use the absolute trading volume of institutional investors, finding very similar results. As expected, we also find a negative coefficient when using the lagged share of retail investors in column (5). A legitimate concern that might arise is that public blockchain data is not representative of the whole crypto market, given the existence of centralized exchanges. We argue that this is not the case as we reach the same conclusions when we use reporting data from Coinbase in column (6).¹⁷ Furthermore, to ensure that our results are not influenced by the construction of factors, we estimate the specification of Table 2 by using the S&P 500 index and the Bitcoin price and find the same results (see Table B.1 in Appendix B).¹⁸

¹⁶Importantly, our findings are robust to using non-smoothed series, which indicates that possible autocorrelation of the trading share is not driving the results.

¹⁷Although Coinbase data is at a quarterly frequency, we still find significant results at the daily level.

¹⁸Our findings are also not confined to the relationship between the crypto factor and equities; for instance, in Table B.2 we show a similar role for institutional investors in explaining the relationship between the crypto factor and global commodities prices.

Table 2: The Role of Institutional Investors in the Correlation between Crypto and Equity Factors

	Δ Crypto Factor _t					
	(1)	(2)	(3)	(4)	(5)	(6)
Δ Global Equity Factor (General) _t	0.149*** (4.86)	0.099*** (3.95)	0.097*** (3.81)	0.101*** (3.93)	0.075*** (3.34)	0.177*** (5.04)
Δ Global Equity Factor (General) _t # Share of Institutionals _{t-1}		0.111*** (3.74)				
Δ Global Equity Factor (General) _t # Share of Institutionals (CEX/DEX) _{t-1}			0.102*** (3.33)			
Δ Global Equity Factor (General) _t # Volume of Institutionals _{t-1}				0.133*** (4.38)		
Δ Global Equity Factor (General) _t # Share Retailers _{t-1}					-0.150*** (-3.80)	
Δ Global Equity Factor (General) _t # Share Inst. Coinbase _{t-1}						0.063*** (2.78)
Controls	YES	YES	YES	YES	YES	YES
Year-Week Fixed Effects	YES	YES	YES	YES	YES	YES
Obs.	1986	1832	1832	1832	1832	1502
R ²	0.053	0.062	0.059	0.078	0.064	0.065

Notes: Variables are standardized; *t*-statistics in parentheses. Controls (VIX, 90-day Bitcoin variance), the lagged levels of the institutional-participation variables interacted in each column, and year-week fixed effects are included but not reported. Daily data, January 2018–March 2026, with each column bounded by the coverage of its institutional-participation measure; the institutional-participation measures are from Chainalysis (on-chain) and Coinbase (off-chain), as labeled. *, **, *** denote significance at the 10%, 5%, and 1% levels.

3.4 The Introduction of Spot Bitcoin ETFs

In this subsection, we complement the previous evidence with a difference-in-differences approach. We exploit the Securities and Exchange Commission’s (SEC) approval of Bitcoin Exchange-Traded Funds (ETFs) in January 2024, which lowered the cost to institutions of gaining exposure to Bitcoin, relative to the cost of gaining exposure to other crypto assets.¹⁹ While some repricing occurred at earlier milestones—such as court decisions and SEC signaling—the approval itself was not fully anticipated: Bitcoin prices jumped in response to an unauthorized approval tweet on the SEC’s official X account and again at the real approval, and within a ± 30 -day window of the approval the Bitcoin price moved by roughly 19% between its low and its high.²⁰ To control for aggregate trends

¹⁹Specifically, Bitcoin ETFs provided a regulated and compliant means for institutions to invest in Bitcoin, and also simplified the investment process since ETF purchases do not require crypto-specific elements such as a crypto wallet or private keys and can be executed through existing trading software. For industry perspectives, see, for instance, TrustWallet (link) and Investor Relations Magazine (link).

²⁰For details, see, for instance, “False Tweet From US Regulator Briefly Sends Bitcoin Soaring” by Agence France Presse on Barron’s (link). For completeness, in Figure B.4 in Appendix B, we report the price of the Bitcoin before and after the SEC’s approval of the ETF.

Table 3: Response of the Share of Institutional Investors to Spot Bitcoin ETF Approval in US

	Share of Institutional Investors			
	(1)	(2)	(3)	(4)
Bitcoin # Post BTC ETF	0.0424 (1.31)	0.0714*** (3.40)	0.0973*** (6.71)	0.1169*** (9.45)
Controls	YES	YES	YES	YES
Day FEs	YES	YES	YES	YES
Coin FEs	YES	YES	YES	YES
Event Window	± 15 days	± 30 days	± 60 days	± 90 days
Obs.	42	84	160	246
R^2	0.963	0.969	0.968	0.964

Notes: t -statistics in parentheses, based on robust standard errors. The coin-volatility control is included but not reported. Daily data; event windows are centered on the SEC's approval of the spot Bitcoin ETF (January 10, 2024). The institutional share is the Chainalysis-based share of on-chain trading volume accounted for by institutional-size transfers ($\geq \$1M$), by coin and day. *, **, *** denote significance at the 10%, 5%, and 1% levels.

in the crypto market, we compare the change in the Bitcoin-equities correlation after the approval to the change in the Ether-equities correlation.²¹ Specifically, we run the following difference-in-differences regression:

$$y_{it} = \alpha_i + \alpha_t + \beta \cdot BTC_i \cdot PostAnnouncement_t + \gamma \cdot volatility_{it} + u_{it} \quad (1)$$

where: $i \in \{\text{Bitcoin, Ether}\}$, y_{it} are outcomes by asset and day, BTC_i is a dummy taking value one for $i = \text{Bitcoin}$, and α_i and α_t are crypto asset and day fixed effects. We also control for the volatility of each asset i , but our results are also robust to excluding this control.

We first define the outcome y_{it} to be the share of institutional investors in asset i , measured using our proxy based on transaction sizes as described above. The results are shown in Table 3, for various event windows (the span of t around the approval date). We find that the share of trading by institutional investors does indeed rise for Bitcoin, relative to Ether, after the announcement of Bitcoin ETFs; the effect is significant in all but the narrowest (± 15 -day) event window, and grows with the window length.

²¹We use Ether as the counterfactual as it has the second-largest market capitalization, so constitutes a relatively close comparator for Bitcoin.

Table 4: Response of the Crypto-Equity Correlation to Spot Bitcoin ETF Approval in US

	Corr(Coin, S&P 500)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Bitcoin # Post ETF Ann.	0.337*** (3.35)	0.241*** (3.32)	0.172*** (4.01)	0.126*** (3.66)	0.091** (2.52)	0.146*** (6.06)	0.099*** (3.07)	0.135*** (6.08)
Controls	YES	YES	YES	YES	YES	YES	YES	YES
Day FEs	YES	YES	YES	YES	YES	YES	YES	YES
Coin FEs	YES	YES	YES	YES	YES	YES	YES	YES
Rolling Correlation	7day	7day	10day	10day	15day	15day	20day	20day
Event Window	±15days	±30days	±15days	±30days	±15days	±30days	±15days	±30days
Obs.	42	84	42	84	42	84	42	84
R^2	0.933	0.936	0.971	0.970	0.932	0.946	0.962	0.964

Notes: t -statistics in parentheses, based on robust standard errors. The coin-volatility control is included but not reported. Daily data; event windows are centered on the SEC's approval of the spot Bitcoin ETF (January 10, 2024); correlations are computed on daily log returns of CryptoCompare US-dollar closing prices and the S&P 500. Event windows of ± 60 days and longer are excluded, as their pre-periods overlap the anticipation of the approval at earlier milestones (see Section 3); rolling-correlation windows are capped at 20 days so that post-announcement correlations are measured from post-announcement returns. *, **, *** denote significance at the 10%, 5%, and 1% levels.

We then turn to our main specification, where we define y_{it} to be the correlation between crypto asset i and the S&P 500, $corr_{it}$.²² The estimates are shown in Table 4, for correlation windows of 7 to 20 days and both event windows. The results corroborate our previous findings: the correlation between Bitcoin and equities increases by 0.09–0.34 in the weeks after the approval, relative to the correlation between Ether and equities, consistent with higher institutional participation increasing the synchronization across asset classes.

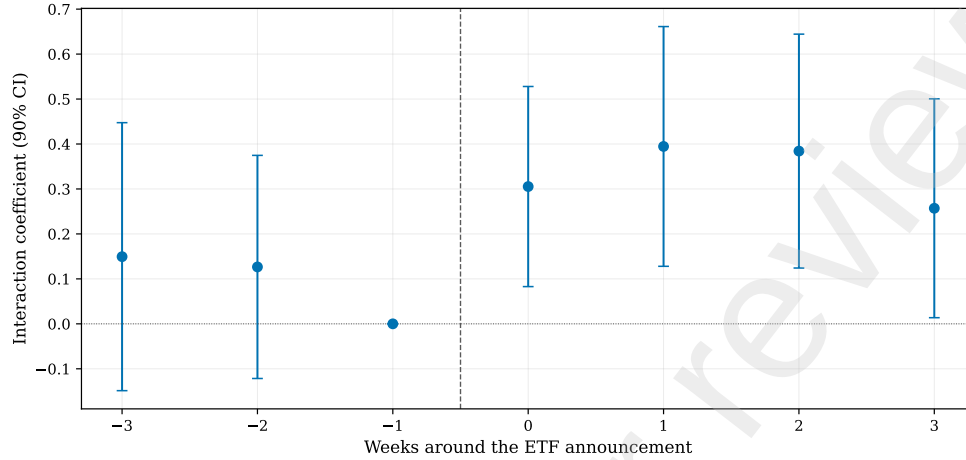
To check for pre-trends, and to examine the dynamics of these results, we additionally run the following specification:

$$corr_{it} = \alpha_i + \alpha_w + \sum_{\substack{\tau=-3 \\ \tau \neq -1}}^3 \beta_\tau \cdot BTC_i \cdot \mathbf{1}(w = \tau) + \gamma \cdot volatility_{it} + \delta \cdot VIX_t + u_{it}. \quad (2)$$

where w stands for event week. Event weeks are defined such that the announcement falls in the middle of week 0, so that week -1 (the reference week) is the last week entirely unaffected by the

²²Correlations are computed on daily log returns; the Bitcoin and Ether price series are the CryptoCompare US-dollar closing prices also used in the construction of the crypto factor, and the S&P 500 series is from FRED; the coin volatility controls are the BitVol® and EthVol volatility indices.

Figure 4: Response to Spot Bitcoin ETF Approval: Dynamics



Notes: This figure plots the coefficients β_τ from regression (2), with correlations calculated using a 10-day rolling window; the reference week is $\tau = -1$. Bars show 90% confidence intervals. Daily data in a ± 3 -week window around the SEC's approval of the spot Bitcoin ETF (January 10, 2024), which falls mid-week 0; CryptoCompare and FRED data.

approval (it ends before the unauthorized approval tweet posted on the SEC's X account on January 9). The results are plotted in Figure 4, using a 10-day rolling correlation. The coefficients are close to zero and insignificant in all pre-announcement weeks, and turn positive from the announcement week onwards: the differential rises by 0.31 in the announcement week itself and remains between 0.26 and 0.39 in the following three weeks, all significant at the 90% level. The immediate response is consistent with the approval carrying genuine news despite the partial anticipation noted above. These dynamics are robust to the choice of the rolling-correlation window (10 to 20 days) and to widening the event window to ± 4 weeks (Figure B.5 in Appendix B).

The identifying assumption of this design is that, absent the ETF approval, the Bitcoin–equity and Ether–equity correlations would have evolved in parallel. The flat pre-announcement coefficients in Figure 4 support this assumption, though it is not directly testable after the event. Since some anticipation of the approval had plausibly been priced in at earlier milestones, the estimated

effect is, if anything, attenuated toward zero.²³

Overall, we find evidence from both a close examination of one special episode (the Bitcoin ETF approval by the SEC) and from correlations over several years (as discussed in the preceding section) that is consistent with institutional participation playing an influential role in crypto markets and their relationship to equities. We next turn to US monetary policy, and investigate the potential role of institutional investors in determining how crypto markets respond.

4 Responses to US Monetary Policy

This section examines how US monetary policy transmits to the crypto cycle. We proceed in three steps. First, we estimate the average response of the crypto factor to high-frequency monetary-policy surprises. Second, we ask whether this response depends on the composition of the crypto investor base, letting the effect vary with the participation of institutional investors. Third, we turn to the mechanism and test whether the transmission is concentrated where the balance-sheet constraints of levered crypto investors bind, as measured by the perpetual-futures funding rate.

4.1 The Impact of Monetary Policy on the Crypto Cycle

We identify US monetary policy shocks using the high-frequency surprises of Acosta et al. (2025), measured over the “monetary event” window (policy statement plus press conference) around FOMC announcements. For each horizon $h = 0, \dots, 7$ trading days, we estimate the local projection

$$Y_{t+h} - Y_{t-1} = \alpha_h + \beta_h \text{MP}_t + \sum_k \gamma_{h,k} X_{k,t-1} + \varepsilon_{t+h}, \quad (3)$$

where Y is the global crypto factor, expressed in standard-deviation units, and MP_t is the surprise in percentage points, set to zero on non-FOMC days. Because the left-hand side is the change in the *level* of the outcome from the day before the announcement ($t-1$) to $t+h$, the coefficient β_h is the

²³Nonetheless, since Bitcoin and Ether also differ in fundamentals, we do not consider this exercise as a perfectly clean natural experiment; rather, we treat it as indicative and complementary to the correlation-based evidence of the preceding section.

cumulative response to a one-percentage-point surprise. The controls X_{t-1} comprise one lag of the daily change in the Treasury 10Y–2Y slope, the broad US dollar index, oil, gold, and the equity and crypto factors—the lagged change of the crypto factor itself controlling for pre-existing trends in the outcome—together with the lagged level of the VIX and the surprise at the previous FOMC meeting, which controls for any autocorrelation of the shocks.²⁴ We use Newey-West standard errors.

The crypto factor falls by 0.41 standard deviations per one-percentage-point surprise on impact (90% CI $[-0.67, -0.14]$), and the response remains negative in the days that follow (Figure 5).²⁵ Since a one-percentage-point surprise is far larger than typical announcement-day surprises, the response to a mean-sized surprise of 0.026 percentage points (the mean absolute surprise across the 71 FOMC events in the sample) is 0.011 standard deviations of the crypto factor, which we consider to be economically meaningful without being implausibly large.²⁶

As a complementary check, we conduct an event study around FOMC announcements. Around each meeting e (announcement day t_0) we compute the crypto factor’s abnormal return $AR_{e,\tau} = r_{t_0+\tau} - \mu_e$, where r_t is the daily change in the (standardized) crypto factor and μ_e is its average daily change over a pre-announcement estimation window (roughly 90 trading days, ending 11 days before the meeting).²⁷ We cumulate abnormal returns from the day before the announcement, $CAR_{e,\tau} = \sum_{s=0}^{\tau} AR_{e,s}$ with $CAR_{e,-1} = 0$, and symmetrically backwards for $\tau < -1$, over a window of ± 7 trading days. To pool meetings of different sign and magnitude into a single “tightening-equivalent” response, we align each meeting to the tightening direction and weight it by the absolute

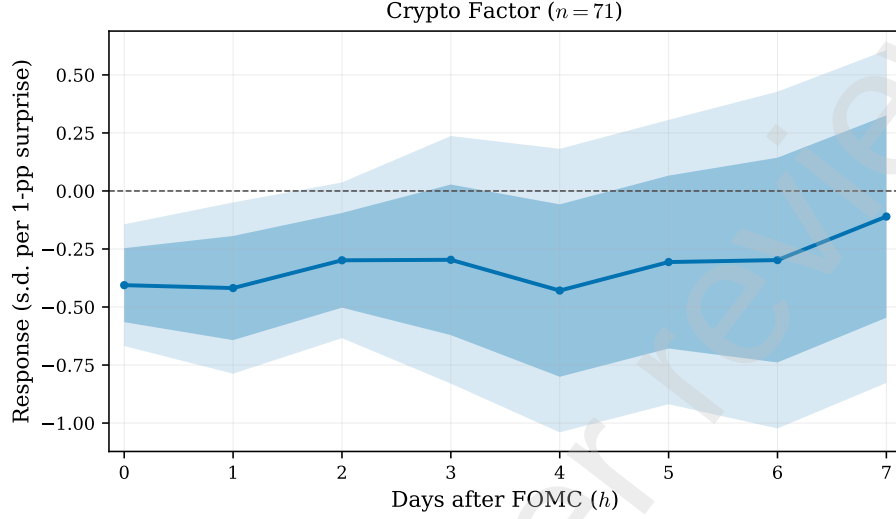
²⁴Because the surprise series is zero except on announcement days, its one-day lag is identically zero on FOMC days; the relevant lag of the shock is therefore the previous meeting’s surprise.

²⁵This finding is robust to the estimation approach. A daily VAR that includes the high-frequency surprise as an internal instrument delivers very similar responses (Figure B.8 in Appendix B). Moreover, because the surprises are estimated, and hence measured with noise, we also re-estimate the baseline by two-stage least squares, using the surprise as an instrument for the announcement-day change in the 1-year or 2-year Treasury yield. The first stages are strong, and the instrumented responses confirm the baseline (Figure B.9 in Appendix B).

²⁶In price terms: one standard deviation of the crypto factor corresponds to approximately 110 log points of the underlying crypto prices (the average across the seven coins of $\text{loading}_i \times \text{sd}(f) \times \text{sd}_i(100 \times \log p_i)$, where $\text{sd}(f)$ is the standard deviation of the estimated factor), so the impact response of -0.41 standard deviations per one-percentage-point surprise corresponds to a decline of roughly 46 log points, i.e., about 37 percent.

²⁷For the first meeting of the sample (January 2018), where fewer than 40 pre-announcement observations of the factor are available, the drift μ_e is set to zero.

Figure 5: The Response of the Crypto Factor to US Monetary Policy



Notes: This figure shows the local-projection impulse response of the (standardized) global crypto factor to a one-percentage-point high-frequency US monetary-policy surprise, estimated from equation (3) on FOMC announcement days, 2018–2026Q1 ($n = 71$). Shaded areas are 68% and 90% confidence bands based on Newey–West (HAC) standard errors. Monetary-policy surprises from Acosta et al. (2025).

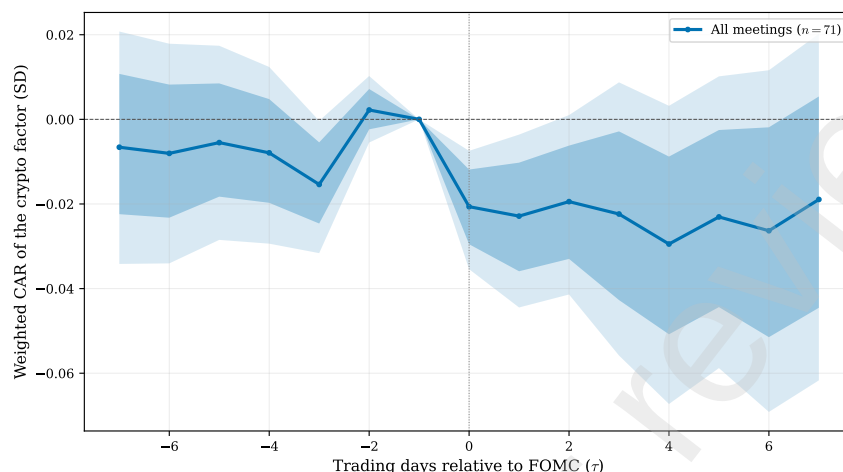
size of its surprise:

$$CAR_{\tau} = \frac{\sum_e |MP_e| [\text{sign}(MP_e) CAR_{e,\tau}]}{\sum_e |MP_e|}, \quad (4)$$

so that a negative value means that a tightening surprise lowers the crypto factor; confidence bands are computed from a nonparametric bootstrap over meetings. We find that a tightening surprise is followed by a significant cumulative decline in the crypto factor (Figure 6), consistent with the local-projection estimates.

Taken together, these results show that the crypto cycle responds to US monetary policy in a similar manner to the global equity cycle. This contrasts starkly with the view that crypto assets are disconnected from the traditional financial system and provide a hedge against market risk. Moreover, the channel we document operates through US monetary policy: comparable high-frequency surprises of the ECB and the Bank of England generate no negative response at any horizon (Figure B.7 in Appendix B). This is consistent with the dollarization of crypto markets:

Figure 6: Event Study around FOMC Announcements



Notes: This figure shows the cumulative abnormal returns of the (standardized) global crypto factor around FOMC announcements, pooled across the 71 meetings in 2018–2026Q1 by aligning each meeting to the tightening direction and weighting it by the absolute size of its monetary-policy surprise, as in equation (4). Shaded areas are 68% and 90% confidence bands from a nonparametric bootstrap over meetings. Monetary-policy surprises from Acosta et al. (2025).

the funding conditions that matter for levered crypto investors are dollar funding conditions, which the Fed sets. Economically, we interpret these responses through the risk-taking channel of monetary policy (e.g., Miranda-Agrippino and Rey, 2020): a monetary tightening makes levered risky positions costlier to sustain and erodes investors’ risk-bearing capacity, prompting them to cut their exposure to risky assets, including crypto. If this is the operative channel, the transmission should depend on the composition of the crypto investor base: the more the marginal crypto investor resembles the levered institutions active in equity markets, the stronger the response to monetary policy. We test this implication next.

4.2 Institutional Investors and Monetary Policy Transmission

To assess how the transmission of US monetary policy depends on the composition of the crypto investor base, we let the response vary with the participation of institutional investors. Specifically,

for each horizon $h = 0, \dots, 7$ on FOMC days, we estimate the regime-interacted local projection

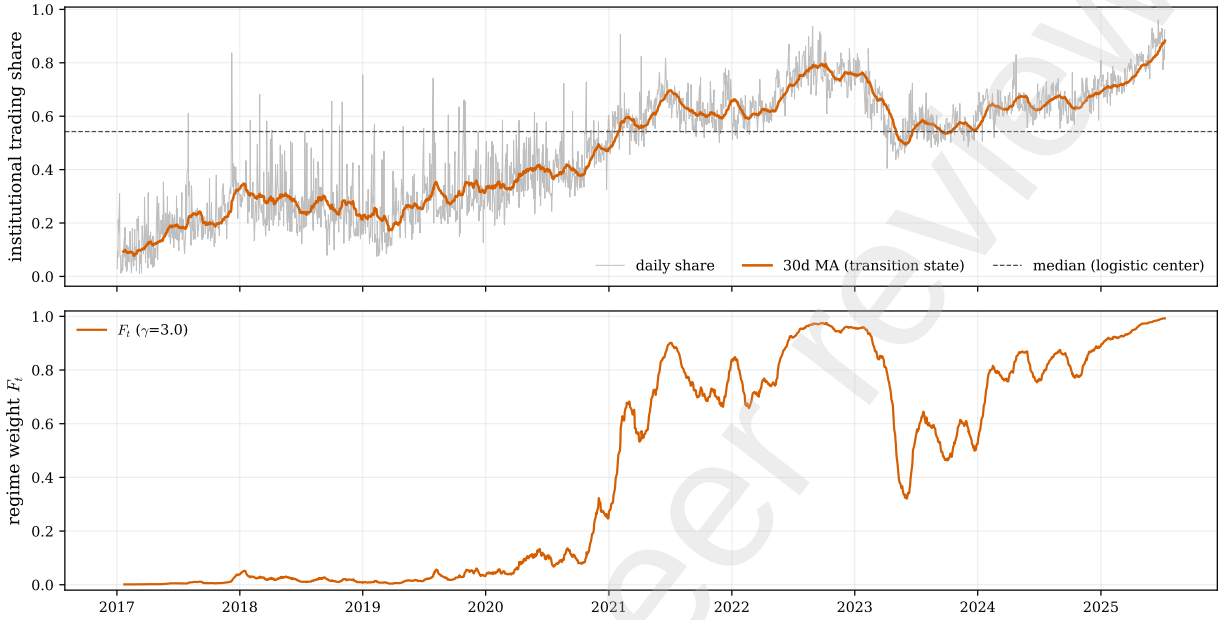
$$Y_{t+h} - Y_{t-1} = \alpha_h + \theta_h F_t + \beta_h^L \text{MP}_t (1 - F_t) + \beta_h^H \text{MP}_t F_t + \phi_h \text{MP}_t D_t^{2020} + \delta_h D_t^{2020} + \sum_k \gamma_{h,k} X_{k,t-1} + \varepsilon_{t+h}, \quad (5)$$

where $F_t = 1/(1 + e^{-\gamma z_{t-1}})$ is a logistic transition of the lagged institutional-investor share: z_{t-1} is the 30-day moving average of the share, standardized around its sample median, and $\gamma = 3$. Thus $F_t \rightarrow 1$ in high-institutional-share periods and $F_t \rightarrow 0$ in low ones, so β_h^H and β_h^L are the monetary-policy responses when institutional participation is high versus low, and $\beta_h^H - \beta_h^L$ is their difference. The institutional-investor share is the Chainalysis-based share of on-chain crypto trading volume accounted for by institutional-size transfers relative to total volume, as described in Section 3. The specification includes a post-2020 dummy (D_t^{2020}) and its interaction with the shock as standard controls, so the regime split is identified over and above a generic post-2020 shift in the monetary-policy response. The shock, the outcome and the control set X_{t-1} are as in equation (3). Figure 7 plots the institutional share, its smoothing, and the implied regime weight F_t .

Figure 8 plots the responses of the crypto factor to a US monetary-policy surprise when institutional participation is low (β_h^L , left panel) and when it is high (β_h^H , middle panel), together with the p -value of the regime difference at each horizon (right panel). The contrast between the two regimes is stark. When institutional participation is low, the response is small and statistically indistinguishable from zero at every horizon. When participation is high, a one-percentage-point tightening surprise lowers the crypto factor by 1.04 standard deviations on impact (90% CI $[-1.73, -0.35]$), and the response remains large and significant at almost every horizon. The difference between the two regimes is itself statistically significant ($p = 0.014$ on impact). In other words, US monetary policy transmits to crypto markets only when institutional investors account for a substantial share of trading, consistent with these investors being the conduit through which changes in the monetary stance reach crypto prices.

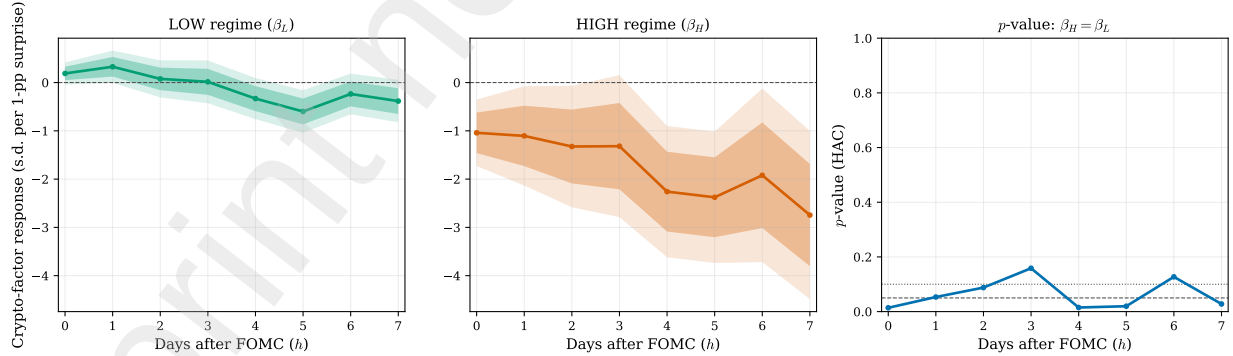
We further subject the regime split to a very demanding specification, in which monetary policy is also allowed to interact with several competing alternatives to the institutional share. Specifically,

Figure 7: The Institutional-Share Regime



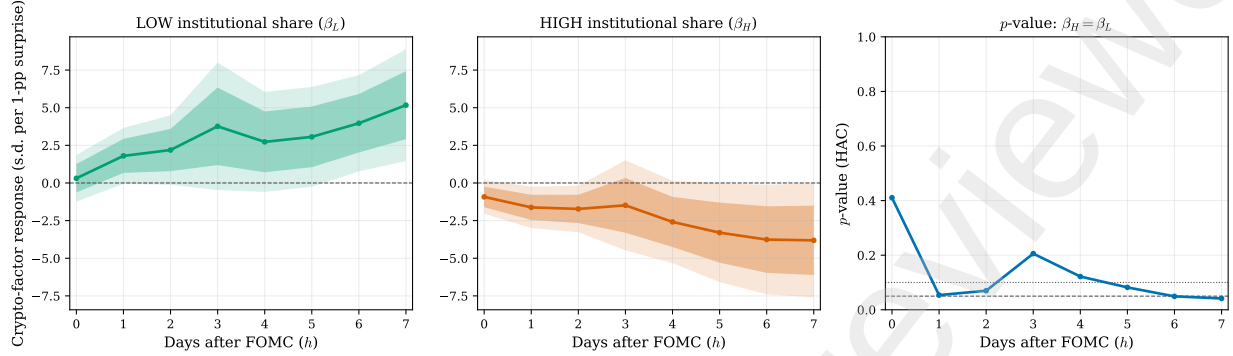
Notes: Top panel: the daily Chainalysis-based institutional trading share, its 30-day moving average (the transition state), and the sample median around which the logistic transition is centered. Bottom panel: the implied regime weight F_t ($\gamma = 3$). Daily data, 2017–July 2025.

Figure 8: Regime-Dependent Responses to US Monetary Policy



Notes: This figure shows the local-projection responses of the (standardized) global crypto factor to a one-percentage-point US monetary-policy surprise in the low-institutional-participation regime (β_L^L , left) and the high-participation regime (β_H^H , middle), estimated from equation (5) on FOMC days, 2018 to July 2025 ($n = 67$), with 68% and 90% HAC confidence bands. The right panel plots the HAC p -value of the regime difference ($H_0: \beta_H^H = \beta_L^L$; dashed line at 0.05, dotted at 0.10). Monetary-policy surprises from Acosta et al. (2025); institutional share from Chainalysis.

Figure 9: Regime-Dependent Responses Controlling for Competing State Variables



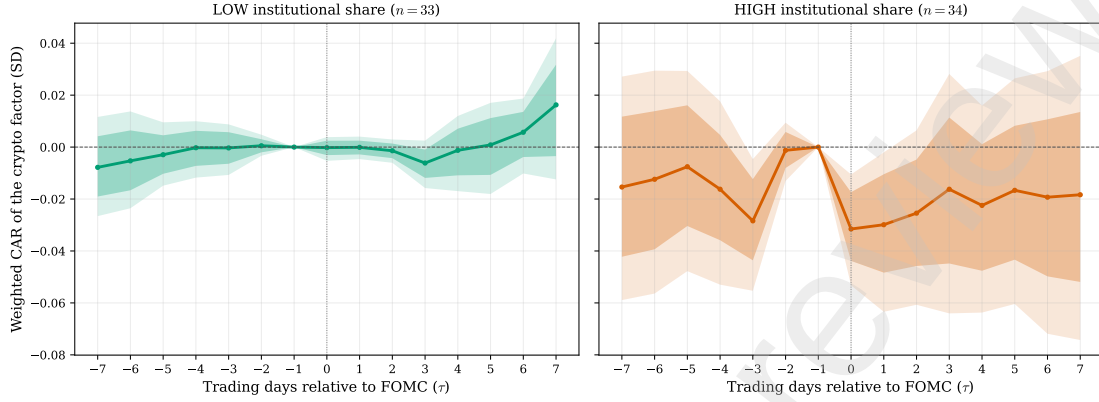
Notes: This figure shows the local-projection responses of the (standardized) global crypto factor to a one-percentage-point US monetary-policy surprise in the low-institutional-participation regime (β_h^L , left) and the high-participation regime (β_h^H , middle), estimated from equation (B.1) in Appendix B, which augments equation (5) with the de-measured interactions of the shock with logistic transitions of five competing state variables (DeFi total value locked, stablecoin market capitalization, the perpetual-futures funding rate, the VIX, and the intermediary capital ratio), entering jointly, together with their de-measured levels. FOMC days, September 2019 to June 2025 ($n = 53$), the sample over which all competing states are available; 68% and 90% HAC confidence bands. The right panel plots the HAC p -value of the regime difference ($H_0: \beta_h^H = \beta_h^L$; dashed line at 0.05, dotted at 0.10). Monetary-policy surprises from Acosta et al. (2025); institutional share from Chainalysis; competing states from DeFiLlama, Binance, CBOE, and He et al. (2017).

we augment equation (5) with the de-measured interactions of the shock with logistic transitions of five competing state variables—DeFi total value locked, the stablecoin market capitalization, the perpetual-futures funding rate (to which we return in Section 4.3), the VIX, and the intermediary capital ratio of He, Kelly, and Manela (2017)—together with their de-measured levels, all entering jointly; Appendix B spells out the specification (equation (B.1)). Because the competing transitions are de-measured, β_h^L and β_h^H retain their interpretation as the monetary-policy responses at the sample average of the competing states. Figure 9 reports the institutional-regime responses from this specification. Even with ten additional parameters and a sample restricted by the availability of the competing states ($n = 53$), the high-participation response remains negative throughout and significant at the 90% level at $h = 1-2$ and $h = 5-7$, and the regime difference remains significant at the same horizons (p between 0.042 and 0.082), although no longer on impact.²⁸

The event-study evidence fully confirms this picture. We compute the sign-aligned, magnitude-

²⁸We also show and discuss the responses associated with the competing states in Appendix B (Figure B.6).

Figure 10: Event Study around FOMC Announcements, by Institutional Regime



Notes: This figure shows the sign-aligned, magnitude-weighted cumulative abnormal returns of the (standardized) global crypto factor around FOMC announcements, as in equation (4), computed separately for meetings occurring in the high- and low-institutional-participation regimes. Shaded areas are 68% and 90% bootstrap confidence bands. FOMC meetings 2018–July 2025; monetary-policy surprises from Acosta et al. (2025).

weighted cumulative abnormal returns of equation (4) separately for meetings in the high- and low-institutional-participation regimes (high versus low F_{t_0}). Exactly as in the local projections, the responses in low-participation periods are flat throughout the event window, whereas in high-participation periods we observe a significant decline (Figure 10). The fact that two distinct empirical approaches—one regression-based, one built from raw abnormal returns—deliver the same regime pattern is reassuring: the results are consistent with the transmission of US monetary policy to crypto markets running through institutional investors, and do not hinge on the specification of the local projections.

These results are robust along several further dimensions, which we report in Appendix B. First, the regime split does not hinge on restricting the coefficients on the controls to be common across regimes: a fully state-interacted specification in the spirit of the state-dependent fiscal-multiplier literature (Ramey and Zubairy, 2018), in which the intercept and every control are also interacted with the regime weights, delivers the same pattern, with an even sharper regime difference on impact (Figure B.10). Second, the split does not depend on how we measure institutional participation: computing the share from exchange-related transactions only delivers very similar

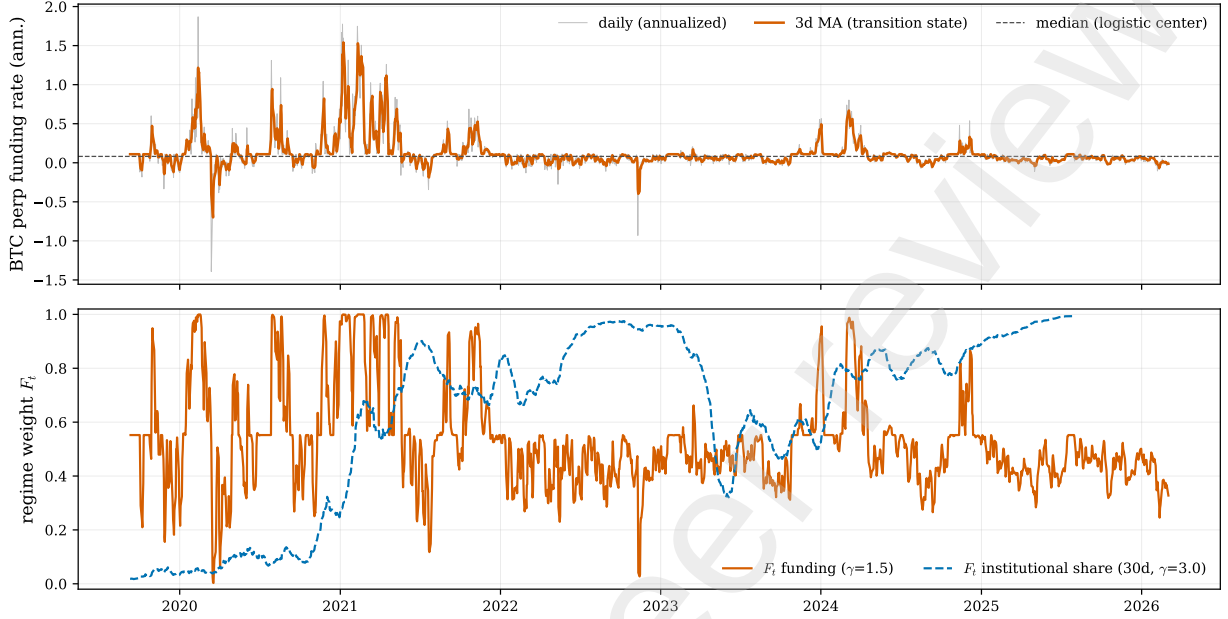
responses (Figure B.11). Third, it does not simply pick up a secular trend: adding a smooth cubic time trend to the controls leaves the results unchanged (Figure B.12). Fourth, it is insensitive to the construction of the regime transition. Sweeping the smoothing window and the curvature of the logistic transition jointly, the high-regime response is negative and the regime difference is significant in every combination (Figure B.13). Varying every remaining ingredient of the transition one at a time (curvature, smoothing, centering, lag of the state, and functional form) produces the same conclusion (Figure B.14). Finally, the results are not an artifact of the factor construction: replacing the crypto factor with the Bitcoin price, which involves no estimation, delivers the same regime pattern in both the local projections (Figure B.15) and the event study (Figure B.16).

4.3 Balance Sheet Channel

To provide additional, complementary evidence of the institutional investor channel, we use the BTC-USDT perpetual-futures funding rate from Binance as a high-frequency proxy for the leverage of crypto investors (Gorton, Klee, Ross, Ross, and Vardoulakis, 2026; Campello et al., 2026). The funding rate is the periodic payment (every eight hours) that anchors the price of perpetual futures—which have no expiry—to the spot price. When demand for leveraged long exposure is high, the perpetual trades above spot and longs pay shorts, so the funding rate rises; when positioning is short or stressed, it turns negative. The funding rate thus measures the shadow cost of levered positioning in crypto markets, i.e., the tightness of the balance-sheet constraints of levered crypto investors.

From the funding rate we build a logistic regime weight $F_t^f = 1/(1 + e^{-\gamma z_{t-1}})$ with $\gamma = 1.5$, where z_{t-1} is the three-trading-day moving average of the annualized funding rate (the nine 8-hour funding prints entering the announcement), lagged one day and standardized around its sample median. Figure 11 plots the funding rate and the implied regime weight, together with the institutional-share transition F^I of Section 4.2: unlike the institutional share, which moves slowly with the secular entry of institutional investors, the funding regime switches frequently throughout

Figure 11: The Funding-Rate State



Notes: Top panel: the annualized BTC-USD perpetual-futures funding rate (Binance), its three-trading-day moving average (the transition state), and the sample median around which the logistic transition is centered. Bottom panel: the implied funding regime weight F_t^f ($\gamma = 1.5$), with the institutional-share transition F_t^I (dashed) for comparison. Daily data, September 2019–March 2026.

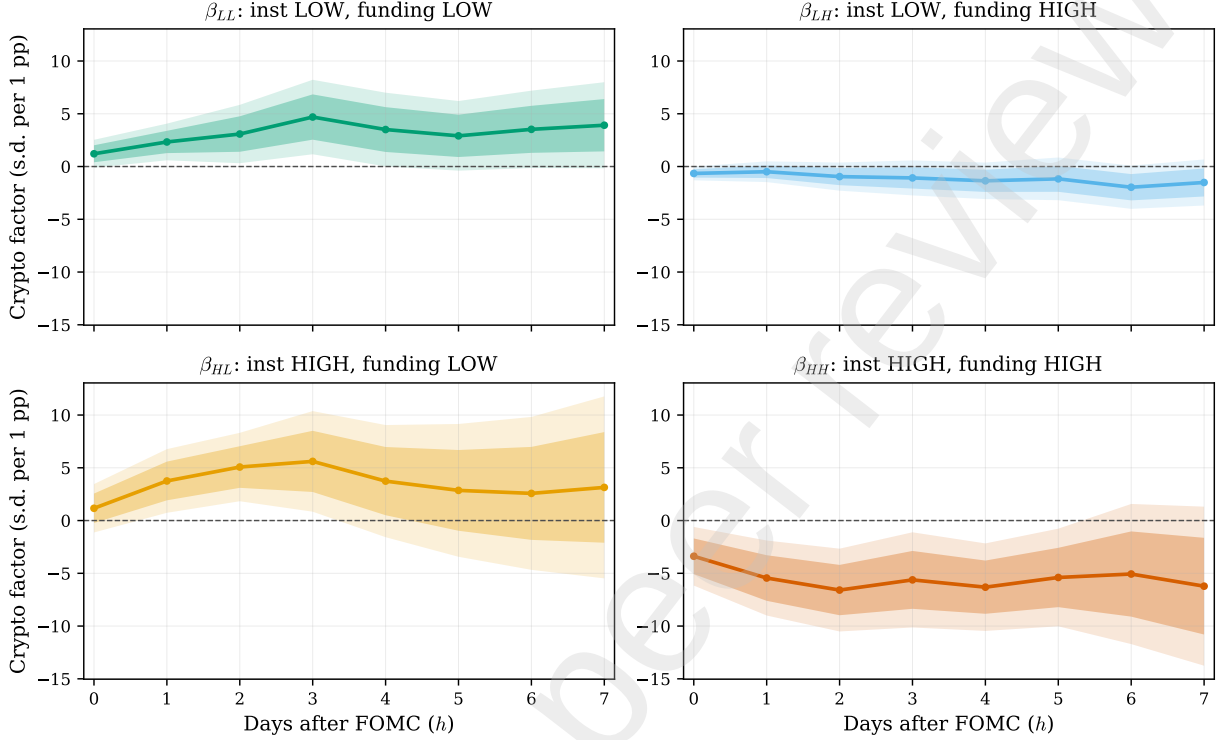
the sample, so the two state variables carry distinct information.

We then interact the two states, splitting the shock into four cells by the institutional-participation regime (F^I , as in equation (5)) and the funding regime (F^f): for horizon $h = 0, \dots, 7$ on FOMC days,

$$Y_{t+h} - Y_{t-1} = \alpha_h + \sum_{r,s \in \{L,H\}} \beta_h^{rs} \text{MP}_t w^r(F_t^I) w^s(F_t^f) + \text{regime levels} + \sum_k \gamma_{h,k} X_{k,t-1} + \varepsilon_{t+h}, \quad (6)$$

with $w^L(F) = 1 - F$ and $w^H(F) = F$, the regime level terms (F_t^I , F_t^f and $F_t^I F_t^f$), and the baseline controls of equation (5), including the post-2020 block. The sample comprises $n = 54$ FOMC events, as the funding rate is available only from September 2019 and the institutional share through July 2025.

Figure 12: Responses by Institutional Participation and Funding Regime

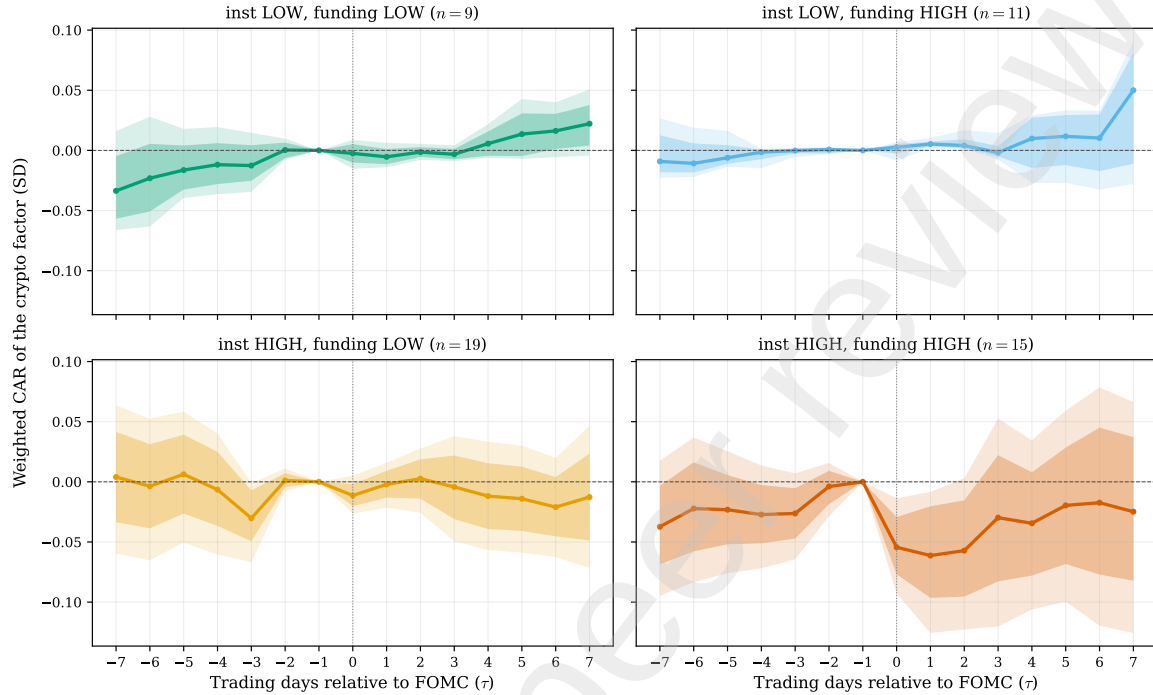


Notes: This figure shows the local-projection responses of the (standardized) global crypto factor to a one-percentage-point US monetary-policy surprise in the four cells of equation (6)—institutional-participation regime $\times$ funding-rate regime—with 68% and 90% HAC confidence bands. FOMC days from September 2019 ($n = 54$). Monetary-policy surprises from Acosta et al. (2025); funding rate from Binance.

We find that the negative response of the crypto factor is concentrated where high leverage coincides with high institutional participation: $\beta_h^{HH} = -3.4$ at $h = 0$ and -6.6 at $h = 2$, significant at the 90% level throughout $h = 0-5$, while none of the other three cells shows a comparable negative response (Figure 12). Within the high-institutional regime, the funding split is significant at the 90% level at $h = 1-4$ (at the 95% level at $h = 1-3$). Institutional participation and the balance-sheet capacity of levered investors thus reinforce each other in the transmission of US monetary policy to crypto markets.

As for the baseline and regime results, we corroborate the triple interaction with an event study. We compute the sign-aligned, magnitude-weighted cumulative abnormal returns of equa-

Figure 13: Event Study around FOMC Announcements, by Institutional Participation and Funding Regime



Notes: This figure shows the sign-aligned, magnitude-weighted cumulative abnormal returns of the (standardized) global crypto factor around FOMC announcements, as in equation (4), computed separately for the four institutional-participation $\times$ funding-regime cells of equation (6). Meetings are assigned to cells by the value of the two regime weights on the announcement day ($F \geq 0.5 = \text{high}$); only the $n = 54$ meetings with both states defined enter ($n = 9, 11, 19$, and 15 meetings in the four cells). Shaded areas are 68% and 90% bootstrap confidence bands. FOMC meetings from September 2019; monetary-policy surprises from Acosta et al. (2025); funding rate from Binance; institutional share from Chainalysis.

tion (4) separately for the four cells, assigning each meeting to a cell according to the two regime weights on the announcement day (high if $F \geq 0.5$, low otherwise, for each state); the sample is the same $n = 54$ meetings with both states defined as in equation (6). We find the same pattern as in the local projections: the cumulative decline is concentrated in the meetings where high institutional participation coincides with high funding rates ($CAR_0 = -0.054$ standard deviations, $CAR_1 = -0.061$, both significant at the 90% level), whereas the other three cells are essentially flat throughout the event window (Figure 13).

These results support our hypothesized transmission mechanism. The funding rate measures

the shadow cost of levered positioning—i.e., how tightly the balance-sheet constraints of levered crypto investors bind. If US monetary policy affected crypto prices purely through other channels, its effect would not depend on this tightness. Instead, the response is concentrated precisely where high institutional participation coincides with binding constraints. This suggests a clear balance-sheet channel: a tightening surprise erodes the risk-bearing capacity of levered investors, and investors who are both present in the market and close to their constraints are forced to unwind positions, amplifying the fall in prices.

Overall, the evidence in this section paints a consistent picture of how US monetary policy reaches the crypto cycle. Monetary policy affects crypto prices, as it does global equities, and the effect is conditional: it operates only when institutional investors account for a substantial share of crypto trading, and, within those periods, it is strongest when the leverage of crypto investors is high. Baseline responses, regime splits, and the double split by participation and funding all point to the same conclusion, whether estimated by local projections or read directly from abnormal returns around FOMC announcements. In this sense, the entry of levered institutional investors has exposed the crypto cycle to a similar risk-taking channel of US monetary policy as that which drives the global financial cycle. The next section formalizes this mechanism in a heterogeneous-agent model.

5 Model

In this section, we provide a stylized model to interpret our empirical results, building on the literature on heterogeneous risk-taking intermediaries (see, for instance: Danielsson, Shin, and Zigrand, 2010; Adrian and Shin, 2014; Miranda-Agrippino and Rey, 2022). We derive an expression for crypto excess returns as a function of the aggregate effective risk aversion Γ_t^c in the crypto market: a wealth-weighted combination of the risk aversion of retail investors and, central to our results, the shadow cost of the balance-sheet constraint of institutional investors. Changes in the composition of the market shift the weight between preferences and constraints in Γ_t^c , and hence crypto prices: the larger the institutional share, the more crypto prices are governed by the tightness of

institutional balance sheets rather than by retail risk appetite. Specifically, changes in the share of institutional investors affect crypto prices and their correlation with global equity prices, as in our empirical results in Section 3. A US monetary contraction disproportionately reduces the wealth of institutional investors, reducing crypto demand and prices, and does so to a greater extent the larger the share of institutional investors in the market, as in Section 4.²⁹

Our framework features two representative heterogeneous agents and two asset classes, namely crypto and equity. Crypto investors c can only invest in crypto markets, whereas institutional investors i can invest in both crypto and equity markets.³⁰ Crypto investors are retail investors that trade using their disposable income and personal savings (see, for instance, Toczynski, 2023). Institutional investors are banks, hedge funds, asset managers and other traditional financial institutions that operate in multiple sectors.³¹ Crypto investors maximize a mean-variance portfolio and can borrow at the US risk-free rate to lever up their positions.³² By contrast, institutional investors are risk-neutral agents that maximize the expected return of their portfolio, given a value-at-risk constraint.³³ The outside option of both agents is to invest in risk-free deposits, which pay zero excess return. Without loss of generality, we interpret the model as having only one crypto asset and only one global stock, which respectively represent the crypto and global equity factors

²⁹Our model does not micro-found crypto returns based on fundamentals; it focuses instead on their relationship to equity returns in equilibrium, and on how that relationship depends on investor wealth shares and financial constraints. For micro-founded models of crypto asset price formation, see, for example, Benigno and Rosa (2023) and Wulfsohn (2025). For a sophisticated model of heterogeneous agents and monetary policy, see Coimbra and Rey (2024).

³⁰We make this simplifying assumption to clarify the exposition of the model, while noting that, empirically, retail investors also have access to the equity market (e.g., through mobile trading apps), but less so than larger institutional investors. Our main results would be unaffected by extending the model to allow both types of investor to participate in both asset classes, with the only constraint that institutional investors are initially under-represented in the crypto market. The key feature of the model is not the difference in the investable universes but the difference in investors' constraints/risk appetite.

³¹Alternatively, one can think of our institutional investors as fundamentalists—diversified investors with lower effective risk aversion whose mandate spans both asset classes—while our retail crypto investors are dedicated crypto speculators.

³²This is a simplifying assumption: arguably, such investors are not granted loans at the risk-free rate but at a rate proportional to it. Indeed, introducing heterogeneous borrowing costs—where borrowing is more expensive for small crypto investors—would support our findings, as the entry of institutional investors would imply an even greater increase in crypto leverage.

³³We also note that a setup with two risk-averse agents, where institutional investors are less risk averse, would generate similar results.

in the empirical analysis.³⁴

Crypto investors: Crypto investors (retailers) maximize a mean-variance portfolio and have a constant risk aversion coefficient σ . They can hold only crypto assets, which pay an excess return R_{t+1}^c . They, therefore, face the following problem:

$$\max_{x_t^c} \mathbb{E}_t(x_t^c R_{t+1}^c) - \frac{\sigma}{2} \text{Var}_t(x_t^c R_{t+1}^c)$$

where x_t^c is the share of wealth w_t^c invested in the crypto asset, while $\mathbb{E}_t$ and Var_t represent the expected value and the variance, respectively. The first order condition is simply $x_t^c = \frac{1}{\sigma} \mathbb{E}_t(R_{t+1}^c) [\text{Var}_t(R_{t+1}^c)]^{-1}$. Thus, crypto investors increase their holdings proportionally with the expected return on the crypto asset and decrease them proportionally with their risk aversion and the variance of their portfolio.

Institutional investors: Institutional investors are risk-neutral agents that maximize the expected returns on their portfolios given a value-at-risk constraint.³⁵ They invest in both crypto assets and equity, and thus choose their holdings of crypto assets to solve the following maximization problem:

$$\max_{x_t^i} \mathbb{E}_t(x_t^i R_{t+1}^c + y_t R_{t+1}^e)$$

$$\text{subject to: } \underbrace{\theta w_t^i \sqrt{\text{Var}_t(x_t^i R_{t+1}^c + y_t R_{t+1}^e)}}_{\text{value-at-risk constraint}} \leq w_t^i$$

where x_t^i is the share of wealth w_t^i invested in crypto and y_t is the share invested in equities, and R_{t+1}^e is the excess return on equity investments. Similarly to Miranda-Agrippino and Rey (2020), the value-at-risk constraint is expressed in terms of a multiple θ of the investors' portfolio. The

³⁴We can equivalently interpret the model as featuring vectors of securities.

³⁵See for instance Adrian and Shin (2014).

first order condition is

$$x_t^i = \frac{1}{2\theta^2\lambda_t} [\mathbb{E}_t(R_{t+1}^c) - 2\theta^2\lambda_t \text{Cov}_t(R_{t+1}^c, R_{t+1}^e)y_t] [\text{Var}_t(R_{t+1}^c)]^{-1}$$

where λ_t is the Lagrange multiplier. Institutional investors' optimal investment in crypto is positively related to the expected payoff of crypto assets and negatively related to (i) the variance of crypto returns, (ii) the covariance of crypto returns with returns on equities, and (iii) the tightness of their financial constraints. We assume that institutional investors are able to take on more risk than the average crypto investor—i.e., $2\theta^2\lambda_t < \sigma$. Thus we have:

Proposition 1: **Crypto excess returns are priced by a wealth-weighted average of retail risk aversion and the balance-sheet constraint of institutional investors.** *The excess return on crypto assets can be rewritten as:*

$$\mathbb{E}_t(R_{t+1}^c) = \Gamma_t^c \text{Var}_t(R_{t+1}^c) s_t + \Gamma_t^c \text{Cov}_t(R_{t+1}^c, R_{t+1}^e) y_t^{\text{tot}} \frac{w_t^i}{w_t^c + w_t^i} \quad (7)$$

where $\Gamma_t^c = (w_t^c + w_t^i) \left[\frac{w_t^c}{\sigma} + \frac{w_t^i}{2\theta^2\lambda_t} \right]^{-1}$ is the aggregate effective risk aversion, s_t is the market value of the crypto asset per unit of total investor wealth $w_t^c + w_t^i$, and y_t^{tot} is the institutional investors' equity portfolio share y_t evaluated in equilibrium.

Intuitively, crypto excess returns in equilibrium must be higher to compensate for their variance, in proportion to the effective risk aversion of the marginal investor, which reflects retail preferences when retail wealth dominates and the tightness of institutional balance sheets when institutional wealth dominates. Similarly, a higher correlation with equities implies lower diversification benefits for institutional investors, increasing the required return on crypto assets in equilibrium, and this matters more the larger the share of wealth held by institutional investors. Note that the institutional contribution to Γ_t^c is $2\theta^2\lambda_t$, the shadow cost of the value-at-risk constraint—a balance-sheet

quantity that rises when the constraint tightens and vanishes when it is slack.

Proposition 2: Equity excess returns are a function of the financial constraints of institutional investors and of their portfolio allocation to crypto assets. The expected excess equity return can be rewritten as the sum of an equity and a crypto component:

$$\mathbb{E}_t(R_{t+1}^e) = 2\theta^2\lambda_t \text{Var}_t(R_{t+1}^e) y_t^{\text{tot}} + 2\theta^2\lambda_t \text{Cov}_t(R_{t+1}^c, R_{t+1}^e) x_t^i \quad (8)$$

Intuitively, in equilibrium investors must be compensated for higher variance or lower diversification benefits in proportion to their financial constraint.

Comparing Equations 7 and 8 then implies three further results, which map onto our empirical findings once the institutional wealth share $\frac{w_t^i}{w_t^c + w_t^i}$ is interpreted as the institutional trading share of Section 3.

First, the composition of the crypto market determines how closely the risk-taking profile of its marginal investor resembles that of the marginal equity investor, and hence how strongly crypto prices co-move with equity prices. The aggregate effective risk aversion Γ_t^c weights retail preferences and the institutional balance-sheet constraint by wealth: a larger institutional share shifts Γ_t^c toward the shadow cost $2\theta^2\lambda_t$ that also prices equities in Proposition 2, and raises the weight $\frac{w_t^i}{w_t^c + w_t^i}$ on the covariance term in Proposition 1. In the extreme case where institutions entirely dominate the crypto market— $\frac{w_t^i}{w_t^c + w_t^i} \rightarrow 1$, so $\Gamma_t^c \rightarrow 2\theta^2\lambda_t$ —preferences drop out of crypto pricing altogether: the effective risk aversion of the marginal crypto investor is entirely a balance-sheet object, and crypto and equity returns differ only in the relative supplies and relative variances of the two assets. This is precisely the pattern in our empirical findings: the rolling crypto-equity correlation tracks the institutional share over time (Figure 3), the lagged institutional share predicts the strength of the co-movement (Table 2), and the correlation of Bitcoin with equities rose, relative to Ether, when the ETF approval increased institutional access to Bitcoin specifically (Tables 3 and 4, Figure 4).

Second, monetary policy moves crypto prices more strongly the larger the institutional presence

in the market. A monetary tightening raises the risk-free rate, which mechanically lowers the prices of both crypto and equities for given expected payoffs. Because institutional investors take on more risk per unit of wealth ($2\theta^2\lambda_t < \sigma$), this common decline erodes their wealth disproportionately, consistent with evidence that more levered agents are more sensitive to the economic cycle (Adrian and Shin, 2014; Coimbra et al., 2022). The wealth reallocation shifts the weight in Γ_t^c from the institutional term $2\theta^2\lambda_t$ toward the higher retail risk aversion σ : the aggregate effective risk aversion of the crypto market rises, so the required excess return on crypto rises and its price falls further. Without institutional participation this second-round effect is absent: $\Gamma_t^c = \sigma$ is constant, so required excess returns do not move and a tightening lowers crypto prices through discounting alone. The amplification grows with the institutional share, as in the regime-dependent local projections of Figure 8 and the regime-split event study of Figure 10.

Third, the channel operates only when the balance-sheet constraint binds. If the constraint is slack ($\lambda_t = 0$), nothing limits institutional positions: any asset offering a positive expected excess return attracts unbounded demand, so in equilibrium expected excess returns on both assets are zero irrespective of the wealth distribution. A rise in the risk-free rate then lowers both prices through discounting alone, with no amplification. When the constraint binds—in periods of high funding rates—balance-sheet capacity is scarce, expected excess returns are positive, and the wealth reallocation raises Γ_t^c : the channel is active. The response of crypto prices to monetary policy is therefore concentrated where high institutional participation coincides with tight leverage, as in the funding-rate results of Figures 12 and 13.

6 Conclusion

Our study highlights the crucial role of institutional investors in the synchronization between crypto and global equity markets and in the transmission of US monetary policy. We provide evidence that the participation of institutional investors in crypto markets significantly drives the correlation between crypto and global equity markets. Furthermore, we show that US monetary policy exerts a

significant influence on the crypto cycle, analogous to its impact on global equity markets, but only when institutional participation is substantial. Within such periods, the response is concentrated where the leverage of crypto investors, measured by the perpetual-futures funding rate, is high. We argue, both empirically and theoretically, that these patterns share a single explanation: large levered investors holding both asset classes create a direct link between crypto and equity markets, so that shocks transmit to crypto prices through the balance-sheet constraints of these investors.

Policymakers should closely monitor the crypto investor base, as higher institutional participation calls for comprehensive regulatory scrutiny; this concern is echoed in recent policy discussions on the connections between crypto and traditional finance (e.g., Bank for International Settlements, 2023; International Monetary Fund, 2023). The growing presence of institutional investors in crypto markets could amplify spillover risks between the two asset classes, and our results indicate that such risks are greatest when high institutional participation is combined with high leverage. Policymakers should therefore consider strategies that enhance transparency and, in particular, monitor the leverage and liquidity of institutions active in crypto markets.

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References

- Aastveit, Knut Are, Gisle James Natvik, and Sergio Sola, 2017, Economic Uncertainty and the Influence of Monetary Policy, *Journal of International Money and Finance* 76, 50–67, <https://doi.org/10.1016/j.jimonfin.2017.05.003>.
- Acosta, Miguel, Andrea Ajello, Michael D. Bauer, Francesca Loria, and Silvia Miranda-Agrippino, 2025, Financial Market Effects of FOMC Communication: Evidence from a New Event-Study Database, Working Paper 2025-30, Federal Reserve Bank of San Francisco, U.S. Monetary Policy Event-Study Database; <https://doi.org/10.24148/wp2025-30>.
- Adrian, Tobias, Tara Iyer, and Mahvash S. Qureshi, 2022, Crypto Prices Move More in Sync With Stocks, Posing New Risks to Financial Stability, IMF Blog, <https://www.imf.org/en/blogs/articles/2022/01/11/crypto-prices-move-more-in-sync-with-stocks-posing-new-risks>.
- Adrian, Tobias, and Hyun Song Shin, 2014, Procyclical Leverage and Value-at-Risk, *Review of Financial Studies* 27, 373–403, <https://doi.org/10.1093/rfs/hht068>.
- Aiello, Darren, Scott R. Baker, Tetyana Balyuk, Marco Di Maggio, Mark J. Johnson, and Jason D. Kotter, 2023, Who Invests in Crypto? Wealth, Financial Constraints, and Risk Attitudes, Working Paper 31856, National Bureau of Economic Research, <https://doi.org/10.3386/w31856>.
- Altavilla, Carlo, Luca Brugnolini, Refet S. Gürkaynak, Roberto Motto, and Giuseppe Ragusa, 2019, Measuring euro area monetary policy, *Journal of Monetary Economics* 108, 162–179, <https://doi.org/10.1016/j.jmoneco.2019.08.016>.
- Auer, Raphael, Marc Farag, Ulf Lewrick, Lovrenc Orazem, and Markus Zoss, 2022, Banking in the shadow of Bitcoin? The institutional adoption of cryptocurrencies, BIS Working Papers 1013, Bank for International Settlements, <https://www.bis.org/publ/work1013.htm>.

- Auer, Raphael, and David Tercero-Lucas, 2022, Distrust or speculation? The socioeconomic drivers of U.S. cryptocurrency investments, *Journal of Financial Stability* 62, 101066, <https://doi.org/10.1016/j.jfs.2022.101066>.
- Bai, Jushan, and Serena Ng, 2002, Determining the Number of Factors in Approximate Factor Models, *Econometrica* 70, 191–221, <https://doi.org/10.1111/1468-0262.00273>.
- Bank for International Settlements, 2023, The Crypto Ecosystem: Key Elements and Risks, Report submitted to the G20 finance ministers and central bank governors, Bank for International Settlements, <https://www.bis.org/publ/othp72.htm>.
- Baur, Dirk G., and Thomas Dimpfl, 2018, Asymmetric Volatility in Cryptocurrencies, *Economics Letters* 173, 148–151, <https://doi.org/10.1016/j.econlet.2018.10.008>.
- Benetton, Matteo, and Giovanni Compiani, 2024, Investors' Beliefs and Cryptocurrency Prices, *Review of Asset Pricing Studies* 14, 197–236, <https://doi.org/10.1093/rapstu/raad015>.
- Benigno, Gianluca, and Carlo Rosa, 2023, The Bitcoin–Macro Disconnect, Staff Reports 1052, Federal Reserve Bank of New York, https://www.newyorkfed.org/research/staff_reports/sr1052.
- Binance, 2017, BNB Whitepaper: Binance Exchange, Whitepaper, Archived official document, binance.com; https://web.archive.org/web/20180527153212/https://www.binance.com/resources/ico/Binance_WhitePaper_en.pdf.
- Braun, Robin, Silvia Miranda-Agrippino, and Tuli Saha, 2025, Measuring monetary policy in the UK: The UK Monetary Policy Event-Study Database, *Journal of Monetary Economics* 149, 103645, <https://doi.org/10.1016/j.jmoneco.2024.103645>.
- Buterin, Vitalik, 2014, Ethereum: A Next-Generation Smart Contract and Decentralized Application Platform, Whitepaper, <https://ethereum.org/en/whitepaper/>.

- Campello, Murillo, Angela Gallo, Lira Mota, and Tammaro Terracciano, 2026, Demand for Safety in the Crypto Ecosystem, Working Paper 35557, National Bureau of Economic Research, <http://www.nber.org/papers/w35557>.
- Coimbra, Nuno, Daisoon Kim, and Hélène Rey, 2022, Central Bank Policy and the concentration of risk: Empirical estimates, *Journal of Monetary Economics* 125, 182–198, <https://doi.org/10.1016/j.jmoneco.2021.08.002>.
- Coimbra, Nuno, and Hélène Rey, 2024, Financial Cycles with Heterogeneous Intermediaries, *Review of Economic Studies* 91, 817–857, <https://doi.org/10.1093/restud/rdad039>.
- Corbet, Shaen, Charles Larkin, Brian M. Lucey, Andrew Meegan, and Larisa Yarovaya, 2020, The impact of macroeconomic news on Bitcoin returns, *The European Journal of Finance* 26, 1396–1416, <https://doi.org/10.1080/1351847X.2020.1737168>.
- Cornelli, Giulio, Sebastian Doerr, Jon Frost, and Leonardo Gambacorta, 2023, Crypto shocks and retail losses, BIS Bulletin 69, Bank for International Settlements, <https://www.bis.org/publ/bisbull69.htm>.
- Danielsson, Jon, Hyun Song Shin, and Jean-Pierre Zigrand, 2010, Risk Appetite and Endogenous Risk, FMG Discussion Paper 647, Financial Markets Group, London School of Economics, <https://www.fmg.ac.uk/sites/default/files/publications/DP647.pdf>.
- Didisheim, Antoine, Martina Fraschini, and Luciano Somoza, 2022, The End of the Crypto-Diversification Myth, Swiss Finance Institute Research Paper 22-53, Swiss Finance Institute, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4138159.
- Elsayed, Ahmed H., and Ricardo M. Sousa, 2024, International monetary policy and cryptocurrency markets: dynamic and spillover effects, *The European Journal of Finance* 30, 1855–1875, <https://doi.org/10.1080/1351847X.2022.2068375>.

- Forbes, Kristin J., and Roberto Rigobon, 2002, No Contagion, Only Interdependence: Measuring Stock Market Comovements, *The Journal of Finance* 57, 2223–2261, <https://doi.org/10.1111/0022-1082.00494>.
- Forni, Mario, Marc Hallin, Marco Lippi, and Lucrezia Reichlin, 2000, The Generalized Dynamic-Factor Model: Identification and Estimation, *The Review of Economics and Statistics* 82, 540–554, <https://doi.org/10.1162/003465300559037>.
- Geweke, John, 1977, The dynamic factor analysis of economic time series, in Dennis J. Aigner, and Arthur S. Goldberger, eds., *Latent Variables in Socio-Economic Models*, 365–383 (North-Holland, Amsterdam).
- Gorton, Gary B., Elizabeth C. Klee, Chase P. Ross, Sharon Y. Ross, and Alexandros P. Vardoulakis, 2026, Leverage and Stablecoin Pegs, *Journal of Financial and Quantitative Analysis* 61, 99–136, <https://doi.org/10.1017/S0022109025000134>.
- Gourinchas, Pierre-Olivier, Hélène Rey, and Nicolas Govillot, 2010, Exorbitant Privilege and Exorbitant Duty, IMES Discussion Paper 10-E-20, Institute for Monetary and Economic Studies, Bank of Japan, <https://www.imes.boj.or.jp/research/papers/english/10-E-20.pdf>.
- Hackethal, Andreas, Tobin Hanspal, Dominique M. Lammer, and Kevin Rink, 2022, The Characteristics and Portfolio Behavior of Bitcoin Investors: Evidence from Indirect Cryptocurrency Investments, *Review of Finance* 26, 855–898, <https://doi.org/10.1093/rof/rfab034>.
- He, Zhiguo, Bryan Kelly, and Asaf Manela, 2017, Intermediary asset pricing: New evidence from many asset classes, *Journal of Financial Economics* 126, 1–35.
- International Monetary Fund, 2023, Elements of Effective Policies for Crypto Assets, IMF Policy Paper 2023/004, International Monetary Fund, <https://www.imf.org/en/Publications/Policy-Papers/Issues/2023/02/23/Elements-of-Effective-Policies-for-Crypto-Assets-530092>.

- Iyer, Tara, 2022, Cryptic Connections: Spillovers between Crypto and Equity Markets, Global Financial Stability Note 2022/001, International Monetary Fund, <https://www.imf.org/en/Publications/global-financial-stability-notes/Issues/2022/01/10/Cryptic-Connections-511776>.
- Karau, Sören, 2023, Monetary policy and Bitcoin, *Journal of International Money and Finance* 137, 102880, <https://doi.org/10.1016/j.jimonfin.2023.102880>.
- Kekre, Rohan, and Moritz Lenel, 2022, Monetary Policy, Redistribution, and Risk Premia, *Econometrica* 90, 2249–2282, <https://doi.org/10.3982/ECTA18014>.
- Liu, Yukun, and Aleh Tsyvinski, 2021, Risks and Returns of Cryptocurrency, *The Review of Financial Studies* 34, 2689–2727, <https://doi.org/10.1093/rfs/hhaa113>.
- Ma, Chaoqun, Yonggang Tian, Shisong Hsiao, and Liurui Deng, 2022, Monetary policy shocks and Bitcoin prices, *Research in International Business and Finance* 62, 101711, <https://doi.org/10.1016/j.ribaf.2022.101711>.
- Miranda-Agrippino, Silvia, and Hélène Rey, 2020, U.S. Monetary Policy and the Global Financial Cycle, *The Review of Economic Studies* 87, 2754–2776, <https://doi.org/10.1093/restud/rdaa019>.
- Miranda-Agrippino, Silvia, and Hélène Rey, 2022, The Global Financial Cycle, in Gita Gopinath, Elhanan Helpman, and Kenneth Rogoff, eds., *Handbook of International Economics*, volume 6, 1–43 (Elsevier), <https://doi.org/10.1016/bs.hesint.2022.02.008>.
- Nakamoto, Satoshi, 2008, Bitcoin: A Peer-to-Peer Electronic Cash System, White paper, <https://bitcoin.org/bitcoin.pdf>.
- Pyo, Sujin, and Jaewook Lee, 2020, Do FOMC and macroeconomic announcements affect Bitcoin prices?, *Finance Research Letters* 37, 101386, <https://doi.org/10.1016/j.fr1.2019.101386>.

A Factors Construction

Dynamic factor models allow us to decompose a set of prices into a common and an idiosyncratic component.³⁶ We start with the daily US-dollar closing prices (obtained from the CryptoCompare API) of the largest crypto assets that were created before January 2018, excluding stablecoins. This leaves us with seven crypto assets, accounting for around 80% of total non-stablecoin crypto market capitalization in June 2022 (CoinMarketCap). These are: Bitcoin, Ethereum, Binance Coin, Ripple, Cardano, Dogecoin, and Tron. We then write this panel of crypto prices p_{it} as a linear combination of an $\text{AR}(q)$ common factor f_t and an asset-specific idiosyncratic disturbance ϵ_{it} (which in turn follows an $\text{AR}(1)$ process):

$$\begin{aligned} p_{it} &= \lambda_i f_t + \epsilon_{it} \\ f_t &= A_1 f_{t-1} + \dots + A_q f_{t-q} + \eta_t \\ \epsilon_{it} &= \rho_i \epsilon_{it-1} + e_{it} \end{aligned} \quad \begin{aligned} & \\ \eta_t &\sim \mathcal{N}(0, \Sigma) \\ e_{it} &\sim \mathcal{N}(0, \sigma_{it}^2) \end{aligned} \quad (\text{A.1})$$

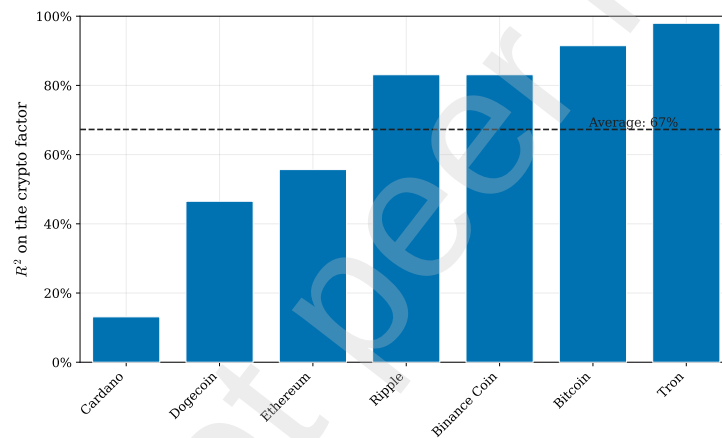
where λ_i is the factor loading of asset i . Estimating this system using maximum likelihood (via the EM algorithm) produces our common factor f_t ; we set $q = 1$. The choice of q is immaterial: information criteria are essentially flat across lag orders, and the estimated factor is virtually identical (pairwise correlations above 0.999) for q between 1 and 5.³⁷ It is also possible to specify multiple factors that affect prices differently. Missing input prices are backfilled by at most five days before estimation, and the saved equity-factor series are linearly interpolated over gaps of up to three trading days. Throughout the empirical analysis, “daily” data refer to a business-day calendar: weekend observations of the continuously traded crypto series are dropped when the panel is merged with traditional financial data.

³⁶For the evolution of the method, see, among others: Geweke (1977); Sargent and Sims (1977); Forni, Hallin, Lippi, and Reichlin (2000); Bai and Ng (2002); Stock and Watson (2002); Miranda-Agrippino and Rey (2020).

³⁷We use the Python class `STATSMODELS/DYNAMICFACTORMQ`. For further information about the model and algorithm, see https://www.statmodels.org/stable/generated/statmodels.tsa.dynamic_factor_mq.DynamicFactorMQ.html.

Figure A.1 reports the R^2 s from univariate regressions of each standardized input price series on the crypto factor. Tron and Bitcoin display the highest R^2 s (98% and 91%), followed by Binance Coin and Ripple (both around 83%); Ethereum and Dogecoin lie somewhat lower (56% and 47%), while Cardano is the clear outlier (13%). The average across assets, marked by the dashed line, is 67%.

Figure A.1: Reverse Regressions



Notes: This figure shows the R^2 s from univariate regressions of each standardized input price series on the crypto factor, as described in Section 3.1; the dashed line marks the average across assets. Daily data, January 2018–March 2026; price data from CryptoCompare.

Table A.1 details the indices utilized for the construction of the global equity factor and its sub-factors. The indices, sourced from Eikon/Thomson Reuters, are categorized by country and sector.³⁸

Table A.1: Equity Eikon RICs by country

Country	Equity Indexes	Tech Indexes	Financial Indexes	Small Caps Indexes
United States	.SPX	.SPLRCT	.SPSY	.SPCY
China	.SSEC	.SZFI	.SZFI	

³⁸Indices whose Eikon coverage does not span the full sample are excluded from factor estimation.

Japan	.JPXNK400			.TOPXS
Germany	.GADXHI	.CXPHX	.CXPVX	
India	.BSESN	.BSETECK	.BSEBANK	
UK	.FTSE	.FTTASX		.FTSC
France	.FCHI	.FRTEC	.FRFIN	.CACCS
Brazil	.BVSP		.TRXFLDBRPFIN	.SMLL
Italy	.FTMIB			.FTITSC
Canada	.GSPTSE	.SPTTTK	.SPTTFS	.SPTSES
Russia	.IRTS		.RTSFN	
South Korea	.KS11	.KRXIT	.KRXBANK	
Australia	.AXJO	.AXIJ	.AXFJ	.AXSO
Spain	.IBEX		.IFNC.MA	.IBEXS
Mexico	.MXX		.MXSE07	.MXXSM
Indonesia	.JKSE			
Turkey	.XU100		.XUMAL	
Netherlands	.AEX		.SXFP	.ASCX
Saudi Arabia	.TASI			
Switzerland	.SSHI	.C9500T	.C8700T	.SSCC
Argentina	.IBG		.TRXFLDARPFIN	
Sweden	.OMXS30			.OMXSSCPI
Poland	.WIG	.COMP	.BNKI	
Belgium	.BFX	.BETEC	.BEFIN	.BELS
Thailand	.SET100	.THTECH	.THFINICIAL	
Austria	.ATX		.TRXFLDATPFIN	
Norway	.OBX			.OSESX
UAE	.DFMGI		.DFMIF	
Nigeria	.NGSEINDEX			
Israel	.TRXFLDILT			
South Africa	.JALSH	.JTECH	.JFINA	.JSMLC
Hong Kong	.HSI	.HSCIIT	.HSCIF	.HSSI
Ireland	.ISEQ			
Denmark	.OMXCBPI			

Singapore	.STI			.FTFSTS
Malaysia	.KLSE	.KLTE	.KLFI	.KLFTSC
Colombia	.COLCAP			
Philippines	.PSI		.PSFI	
Pakistan	.KSE		.TRXFLDPKPFIN	
Chile	.SPCLXIGPA		.TRXFLDCLPFIN	
Finland	.OMXHPI			
Bangladesh	.dMIBD00000P			
Egypt	.EGX30		.TRXFLDEGPFIN	
Vietnam	.VNI			
Portugal	.PSI20	.PTTEC	.PTFIN	
Czech Republic	.PIX			
Romania	.BETI			
Peru	.SPBLPGPT			
New Zealand	.NZ50			.NZSC

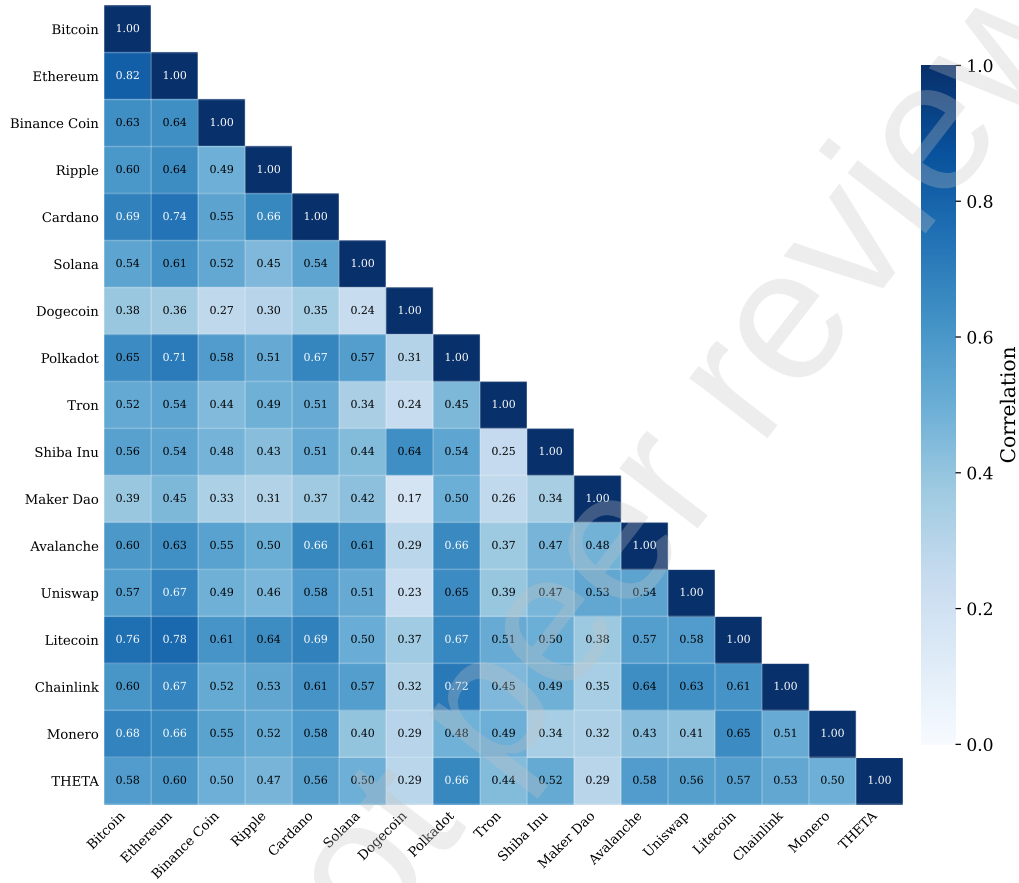
Notes: This table lists the indices used for constructing the global equity factor and each of the equity sub-factors. The selected countries are the fifty largest by GDP. All indices are from Eikon/Thomson Reuters.

B Additional Results and Robustness Tests

B.1 Crypto and Equity Correlations

Figure B.1 reports the pairwise correlations of daily returns among the crypto assets with the largest market capitalization, as discussed in Section 3.1.

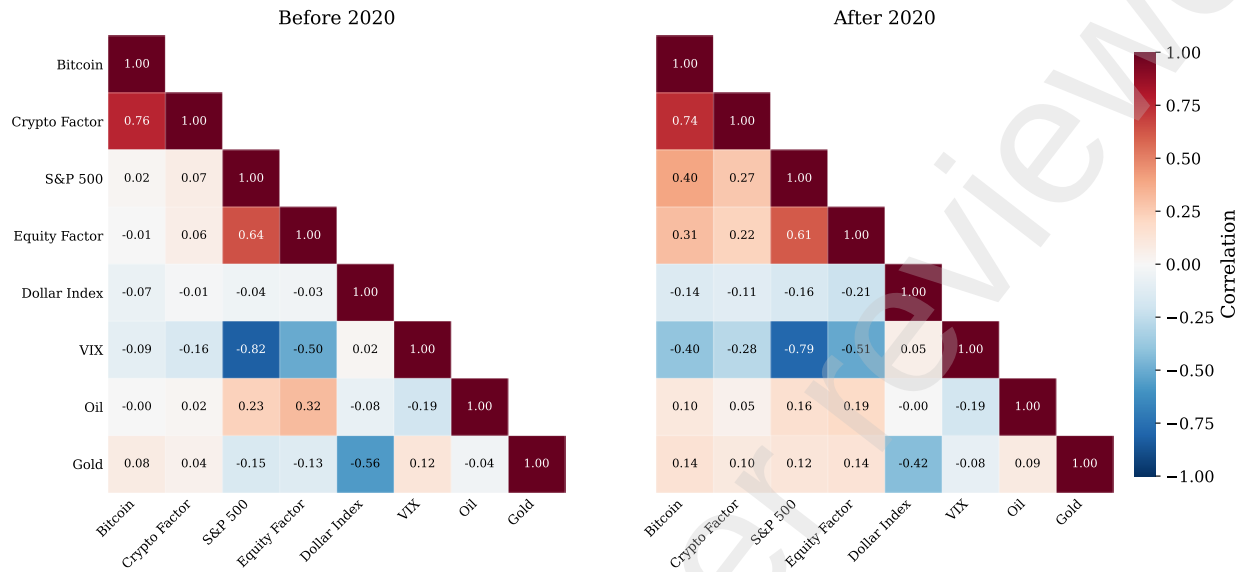
Figure B.1: Correlations among Crypto Assets



Notes: This figure reports the pairwise correlations of daily returns among the crypto assets with the largest market capitalization, over the sample January 2018–March 2026. Price data from CryptoCompare.

Figure B.2 reports the pairwise correlations among crypto, equity and macro-financial variables, before and after 2020, as discussed in Section 3.2.

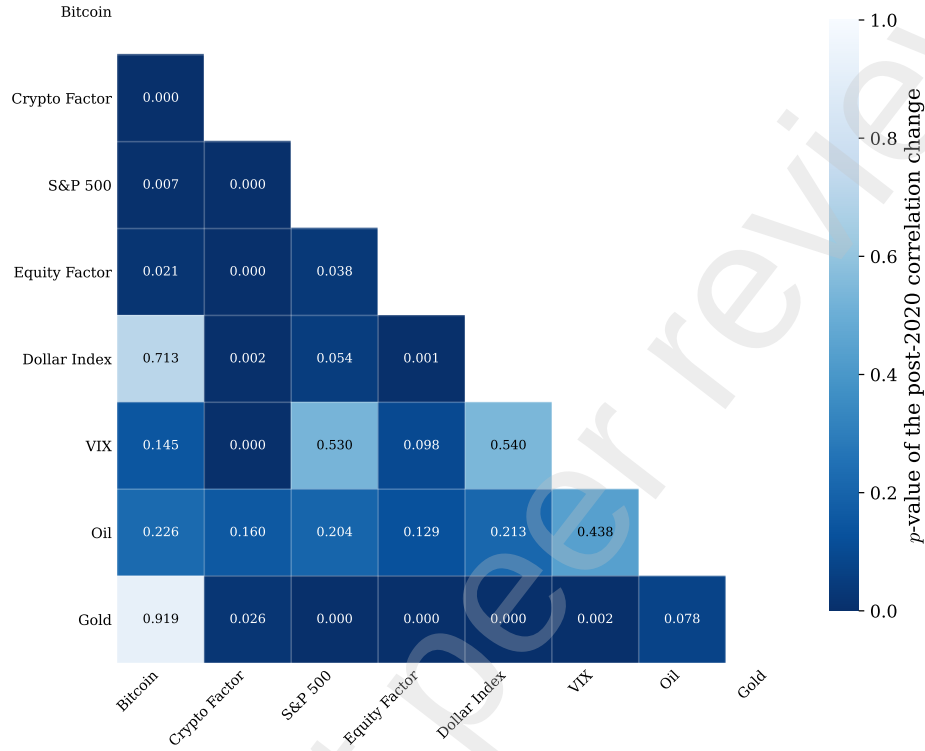
Figure B.2: Correlations Before and After 2020



Notes: This figure reports the pairwise correlations of daily changes among crypto, equity and macro-financial variables, before 2020 (left panel) and from January 2020 onward (right panel). Bitcoin, the S&P 500, the dollar index, oil and gold enter as daily log returns; the (standardized) crypto and equity factors and the VIX enter as daily changes. Crypto prices from CryptoCompare; equity indices from Eikon/Thomson Reuters; macro-financial series from FRED.

Figure B.3 reports the p -values for the differences in correlation before and after the year 2020. The correlations of Bitcoin and of the crypto factor with the S&P 500 change significantly after 2020 (p -values of 0.007 and 0.000, respectively). By contrast, the correlations with oil do not change significantly (p -values of 0.226 and 0.160), and gold shifts only with respect to the crypto factor ($p = 0.026$; $p = 0.919$ for Bitcoin).

Figure B.3: Significance of the Post-2020 Changes in Correlations



Notes: Each cell reports the HC1-robust p -value of β_3 from the regression $y_t = \beta_0 + \beta_1 x_t + \beta_2 D_t + \beta_3 x_t D_t + \varepsilon_t$, where y is the *column* variable, x the *row* variable, and D_t equals one from January 2020—i.e., the significance of the post-2020 change in the slope of y on x . Since the slope of y on x scales with the inverse variance of x , and crypto variance is much larger after 2020, the cells in the Bitcoin and Crypto Factor columns (crypto variable as dependent) are the relevant reading. Bitcoin, the S&P 500, the dollar index, oil and gold enter as daily log returns; the (standardized) crypto and equity factors and the VIX as daily changes; all variables are standardized. Daily data, 2018–2026Q1. Darker cells indicate lower p -values. Crypto prices from CryptoCompare; equity indices from Eikon/Thomson Reuters; macro-financial series from FRED.

Table B.1 re-estimates the specification of Table 2 using the Bitcoin price and the S&P 500 index in place of the crypto and equity factors. The coefficients and their significance levels are consistent with those of Table 2, confirming that the results are not driven by the factor construction.

Table B.1: The Role of Institutional Investors in the Correlation between Bitcoin and S&P 500

	$\Delta \text{Bitcoin}_t$				
	(1)	(2)	(3)	(4)	(5)
$\Delta \text{S\&P } 500_t$	0.337*** (11.51)	0.206*** (9.80)	0.219*** (10.31)	0.174*** (8.31)	0.372*** (12.22)
$\Delta \text{S\&P } 500_t \# \text{ Share of Institutionals}_{t-1}$		0.164*** (6.75)			
$\Delta \text{S\&P } 500_t \# \text{ Volume of Institutionals}_{t-1}$			0.168*** (7.37)		
$\Delta \text{S\&P } 500_t \# \text{ Share Retailers}_{t-1}$				-0.215*** (-6.85)	
$\Delta \text{S\&P } 500_t \# \text{ Share Inst. Coinbase}_{t-1}$					0.118*** (5.53)
Controls	YES	YES	YES	YES	YES
Year-Week Fixed Effects	YES	YES	YES	YES	YES
Obs.	3735	2073	2073	2073	1501
R^2	0.108	0.121	0.132	0.124	0.143

Notes: Variables are standardized; t -statistics in parentheses. Controls (VIX, 90-day Bitcoin variance), the lagged levels of the institutional-participation variables interacted in each column, and year-week fixed effects are included but not reported. Daily data; column (1) is estimated on the full joint Bitcoin–S&P 500 history (October 2010–March 2026), while the remaining columns are bounded by the coverage of the institutional-participation measures, from Chainalysis and Coinbase, as labeled. Bitcoin prices from CryptoCompare; the S&P 500 from Eikon/Thomson Reuters. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table B.2 re-estimates the specification of Table 2 using the MSCI World Commodities Index in place of the equity factor. The role of institutional investors extends to the correlation between the crypto factor and global commodity prices, with results consistent with those for global equities.

Table B.2: The Role of Institutional Investors in the Correlation between Crypto Factor and MSCI World Commodities Index

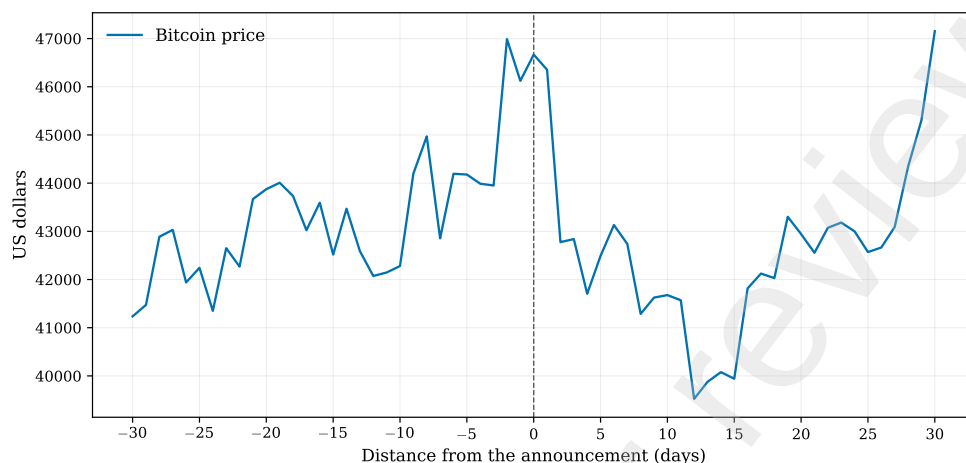
	Δ Crypto Factor _{<i>t</i>}					
	(1)	(2)	(3)	(4)	(5)	(6)
Δ MSCI World Commodities _{<i>t</i>}	0.126*** (5.19)	0.065*** (3.27)	0.066*** (3.17)	0.070*** (3.73)	0.053*** (2.96)	0.143*** (5.27)
Δ MSCI World Commodities _{<i>t</i>} # Share of Institutionals _{<i>t-1</i>}		0.081*** (4.01)				
Δ MSCI World Commodities _{<i>t</i>} # Share of Institutionals (CEX/DEX) _{<i>t-1</i>}			0.075*** (3.65)			
Δ MSCI World Commodities _{<i>t</i>} # Volume of Institutionals _{<i>t-1</i>}				0.087*** (3.44)		
Δ MSCI World Commodities _{<i>t</i>} # Share Retailers _{<i>t-1</i>}					-0.107*** (-3.97)	
Δ MSCI World Commodities _{<i>t</i>} # Share Inst. Coinbase _{<i>t-1</i>}						0.053*** (2.76)
Controls	YES	YES	YES	YES	YES	YES
Year-Week Fixed Effects	YES	YES	YES	YES	YES	YES
Obs.	2080	1916	1916	1916	1916	1577
<i>R</i> ²	0.049	0.049	0.047	0.060	0.050	0.060

Notes: Variables are standardized; *t*-statistics in parentheses. Controls (VIX, 90-day Bitcoin variance), the lagged levels of the institutional-participation variables interacted in each column, and year-week fixed effects are included but not reported. Daily data, January 2018–March 2026, with each column bounded by the coverage of its institutional-participation measure; institutional-participation measures from Chainalysis and Coinbase, as labeled. *, **, *** denote significance at the 10%, 5%, and 1% levels.

B.2 Bitcoin ETF Approval

Figure B.4 plots the Bitcoin price around the SEC's approval of the spot Bitcoin ETF, as discussed in Section 3.

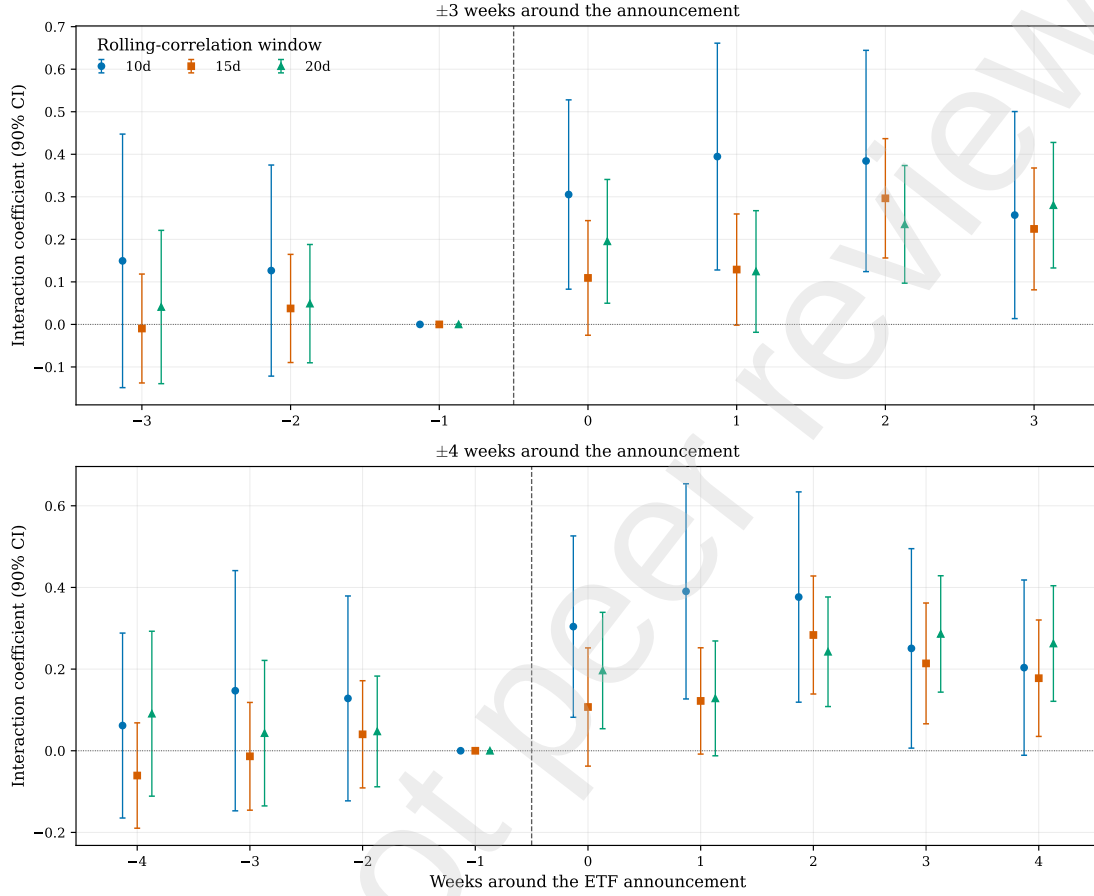
Figure B.4: Bitcoin Price before and after the SEC's ETF Approval



Notes: Bitcoin price in a ± 30 -day window around the SEC's approval of the spot Bitcoin ETF (January 10, 2024). Daily CryptoCompare data.

Figure B.5 reports the weekly event-study coefficients around the ETF announcement of equation (2) for rolling-correlation windows of 10, 15 and 20 days, over both a ± 3 - and a ± 4 -week event window. Rolling windows are capped at 20 days because a w -day rolling correlation dated t is computed from the previous w days of returns, so longer windows would mechanically import the anticipation phase preceding the approval into the event window. Across all specifications, the coefficients are close to zero before the announcement and positive after it, significantly so at most horizons.

Figure B.5: Robustness of the ETF Event Study



Notes: ETF event-study coefficients ($\text{BTC} \times \text{event-week interactions}$, reference = week -1 ; 90% confidence intervals) across rolling-correlation windows of 10 to 20 days, over ± 3 (top panel) and ± 4 (bottom panel) event weeks. The announcement falls mid-week 0, as in Figure 4. Daily CryptoCompare and FRED data around the January 10, 2024 announcement.

B.3 Crypto Response to US Monetary Policy

We begin with the specification behind Figure 9, which subjects the institutional-share regime split to a joint horse race against competing state variables. For each horizon $h = 0, \dots, 7$ on FOMC

days, we estimate

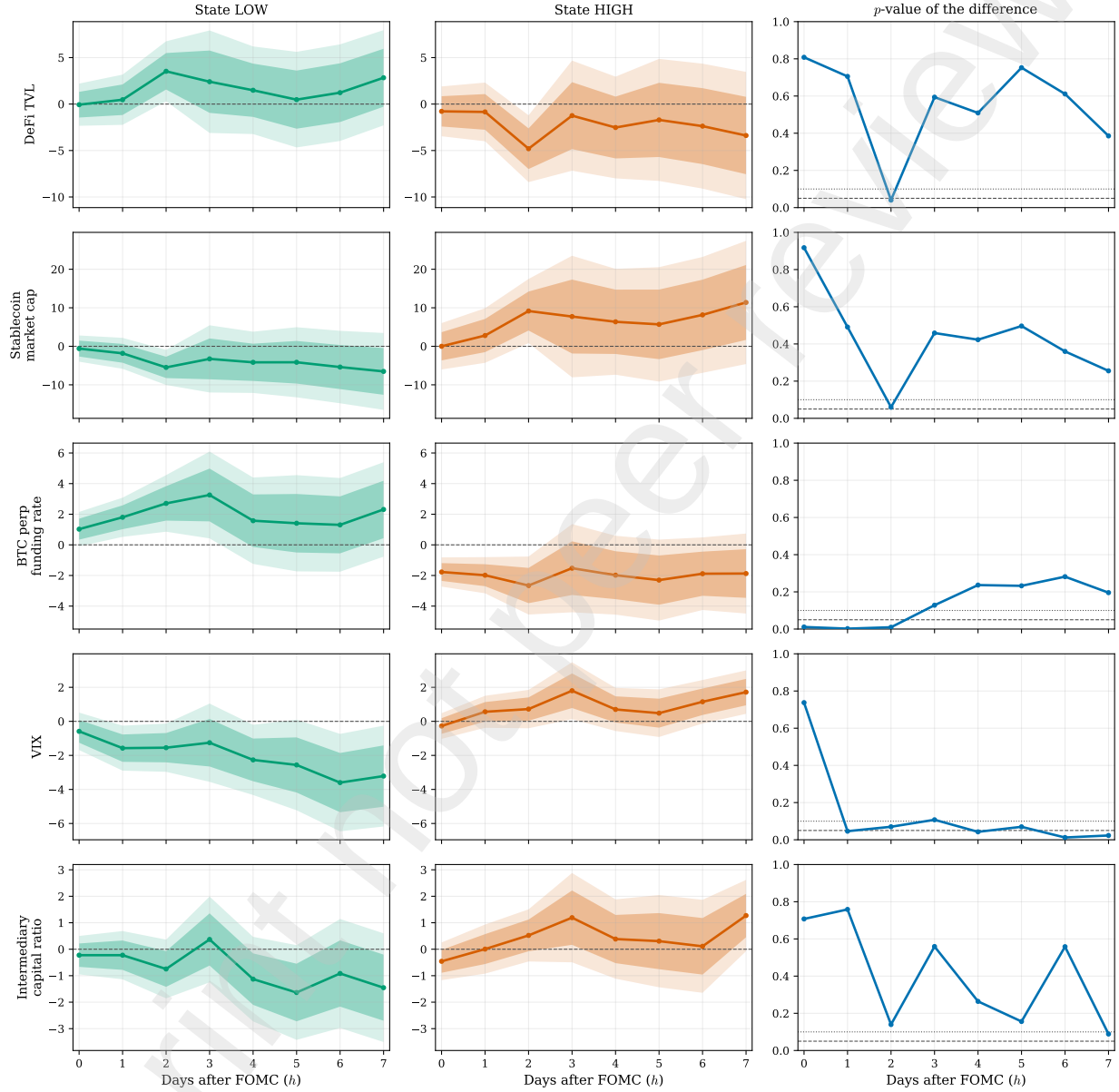
$$\begin{aligned}
Y_{t+h} - Y_{t-1} = & \alpha_h + \theta_h F_t + \beta_h^L \text{MP}_t (1 - F_t) + \beta_h^H \text{MP}_t F_t + \sum_j [b_h^j \text{MP}_t \tilde{F}_t^j + c_h^j \tilde{F}_t^j] \\
& + \phi_h \text{MP}_t D_t^{2020} + \delta_h D_t^{2020} + \sum_k \gamma_{h,k} X_{k,t-1} + \varepsilon_{t+h},
\end{aligned} \tag{B.1}$$

where $\tilde{F}_t^j = F_t^j - \bar{F}^j$ is the de-meaned logistic transition of competing state j , $\bar{F}^j$ is its mean over the impact-horizon estimation sample, and everything else is as in equation (5). The competing states are DeFi total value locked and the total stablecoin market capitalization (from DeFiLlama, in logs), the BTC-USDT perpetual-futures funding rate (three-trading-day moving average, $\gamma = 1.5$), the VIX, and the intermediary capital ratio of He et al. (2017); unless noted, each transition follows the same recipe as the institutional share (30-day moving average, lagged one day, standardized around its sample median, logistic with $\gamma = 3$). This is a very demanding, tightly parameterized specification: all five competing interactions enter jointly, so the regime split must survive every alternative at once, on a sample restricted to the availability of all competing states (September 2019 to June 2025, $n = 53$ FOMC events). The de-meaned levels $\tilde{F}_t^j$ ensure that the interactions do not pick up omitted level effects, and the de-meaning of the transitions implies that at the sample average of the competing states the equation reduces to equation (5), so β_h^L and β_h^H retain their interpretation. Figure 9 in the main text reports the institutional-regime responses.

Figure B.6 reports the responses associated with the competing states: for each state j , the implied response of the crypto factor when that state is low ($F^j = 0$) and high ($F^j = 1$), holding the institutional transition and every other state at its sample mean, together with the p -value of the low/high difference (which equals the test on the interaction coefficient b_h^j). For most of the competing states, the low/high difference is not statistically significant at almost any horizon: DeFi total value locked and the stablecoin market capitalization—whose transitions are strongly correlated with the institutional-share transition (correlations of 0.95 and 0.96)—and the intermediary capital ratio show no stable pattern. Two states deserve comment. For the VIX, the response to

monetary policy is negative and significant at most horizons when the VIX is low, and close to zero when it is high—consistent with the evidence that monetary policy is less powerful in times of high uncertainty (Aastveit, Natvik, and Sola, 2017). For the funding rate, the response is negative and significant at short horizons when the funding rate is high, and the low/high difference is significant at $h = 0-2$: this points to potential complementarities between the balance-sheet state of levered crypto investors and institutional participation, precisely the interaction that the triple-interacted specification of Section 4.3 estimates directly.

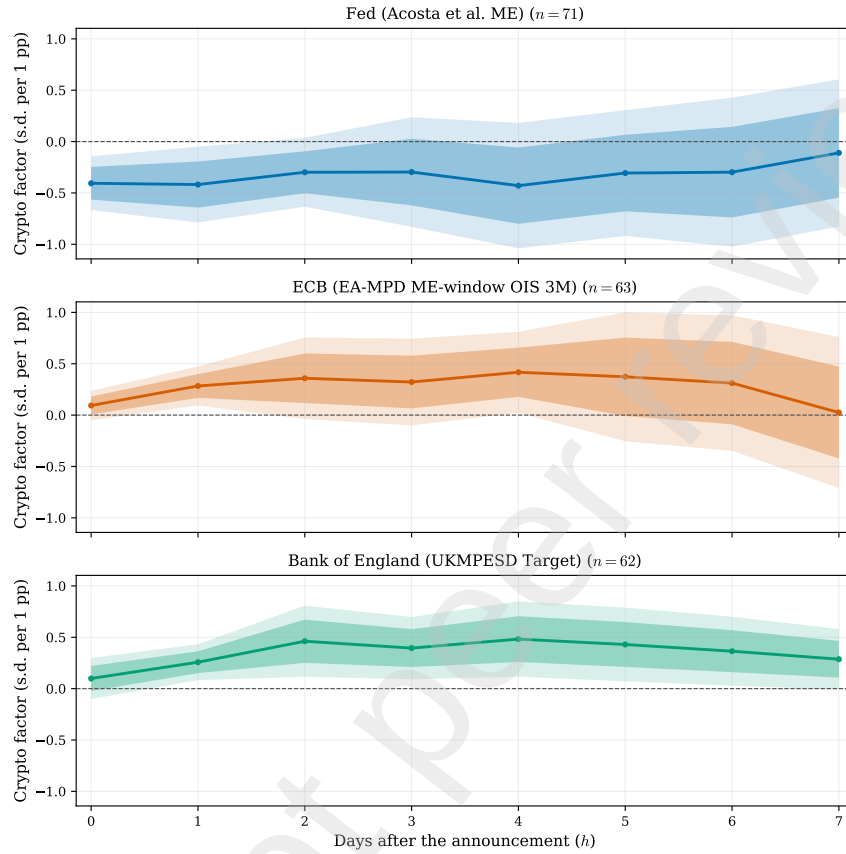
Figure B.6: Horse Race: Responses by Competing State Variables



Notes: Implied local-projection responses of the (standardized) global crypto factor to a one-percentage-point US monetary-policy surprise when each competing state of equation (B.1) is low ($F^j = 0$, left column) and high ($F^j = 1$, middle column), holding the institutional-share transition and every other state at its sample mean, with 68% and 90% delta-method HAC confidence bands. The right column plots the HAC p -value of the low/high difference (dashed line at 0.05, dotted at 0.10). FOMC days, September 2019 to June 2025 ($n = 53$). Monetary-policy surprises from Acosta et al. (2025); institutional share from Chainalysis; competing states from DeFiLlama, Binance, CBOE, and He et al. (2017).

We next examine whether the response of crypto markets is specific to the Federal Reserve or extends to other major central banks. For the ECB, we use the Monetary Event window (press release plus press conference) surprise in the 3-month OIS rate from the Euro Area Monetary Policy Event-Study Database (Altavilla, Brugnolini, Gürkaynak, Motto, and Ragusa, 2019); for the Bank of England, the “Target” factor of the UK Monetary Policy Event-Study Database (Braun, Miranda-Agrippino, and Saha, 2025) on MPC decision days. Each surprise enters the same local projection (3), estimated on that central bank’s own announcement days; all three surprises are in percentage points and have nearly identical standard deviations (≈ 0.03 pp), so the responses per one-percentage-point surprise are directly comparable across central banks. While the crypto factor falls significantly after Fed surprises, ECB and Bank of England surprises generate no negative response at any horizon; if anything, the point estimates are slightly positive (Figure B.7). The channel we document thus operates through US monetary policy.

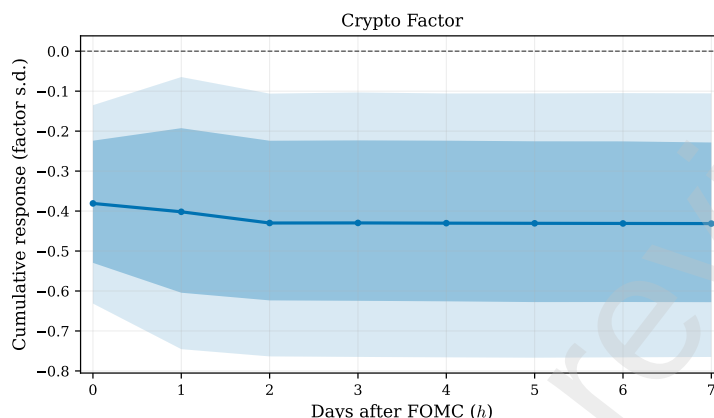
Figure B.7: Fed versus ECB and Bank of England Surprises



Notes: This figure shows the local-projection responses of the (standardized) global crypto factor to a one-percentage-point high-frequency monetary-policy surprise of the Fed (top), the ECB (middle), and the Bank of England (bottom), each estimated on that central bank's own announcement days with the controls of equation (3). Shared y -axis across panels; 68% and 90% HAC confidence bands. Announcement days 2018–2026Q1; surprises from Acosta et al. (2025) (Fed), Altavilla et al. (2019) (ECB), and Braun et al. (2025) (Bank of England).

As a complement to the local projections of Section 4.1, Figure B.8 embeds the high-frequency surprise in a daily VAR: the surprise is added as an endogenous variable ordered first (internal instrument) in an eight-variable VAR(1) with the baseline system variables, identified recursively. The response of the crypto factor is very similar to the local-projection estimate.

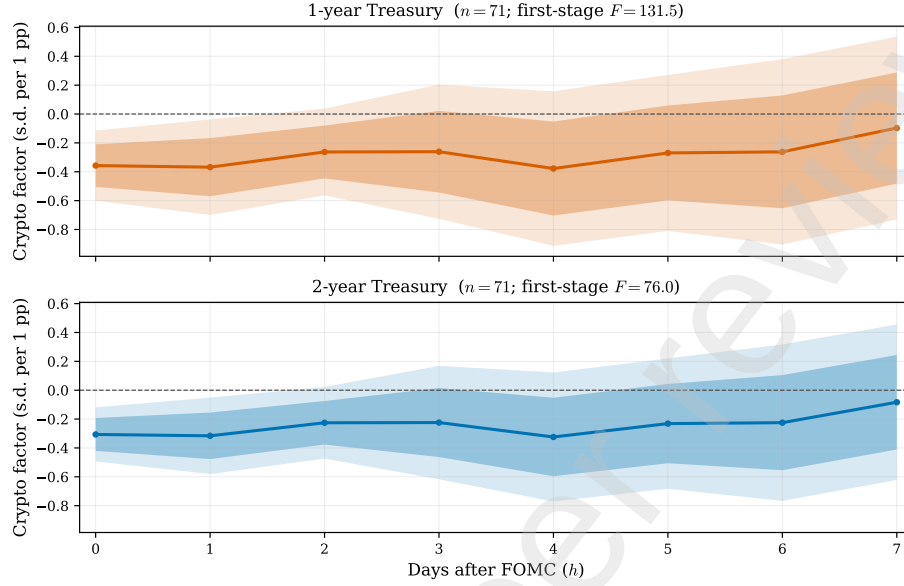
Figure B.8: VAR with the High-Frequency Surprise as Internal Instrument



Notes: Cumulative impulse response of the (standardized) global crypto factor to the high-frequency monetary-policy surprise in a daily VAR(1), with the surprise added as an endogenous variable ordered first and identified recursively. The response is rescaled to a one-percentage-point surprise; daily sample 2018–2026Q1 (71 FOMC events); Monte-Carlo 68% and 90% confidence bands. Monetary-policy surprise from Acosta et al. (2025).

Since the high-frequency surprises are estimated, and hence measured with noise, we also estimate an instrumental-variable version of the baseline local projection, in which the monetary-policy surprise serves as an instrument rather than as a regressor. For each horizon, we estimate equation (3) by two-stage least squares, replacing the surprise with the announcement-day change in a short-maturity Treasury yield—the 1-year and the 2-year—instrumented by the surprise. The coefficients are thus rescaled to the response per one-percentage-point change in the yield, and measurement noise in the surprise no longer attenuates them. The instrument is strong in both cases (robust first-stage F of 131.5 and 76.0), and the responses confirm the baseline: the crypto factor falls by 0.36 (1-year) and 0.31 (2-year) standard deviations per one-percentage-point rise in the yield on impact, both significant at the 90% level on impact and at $h = 1$, with the same profile as the reduced-form estimates (Figure B.9).

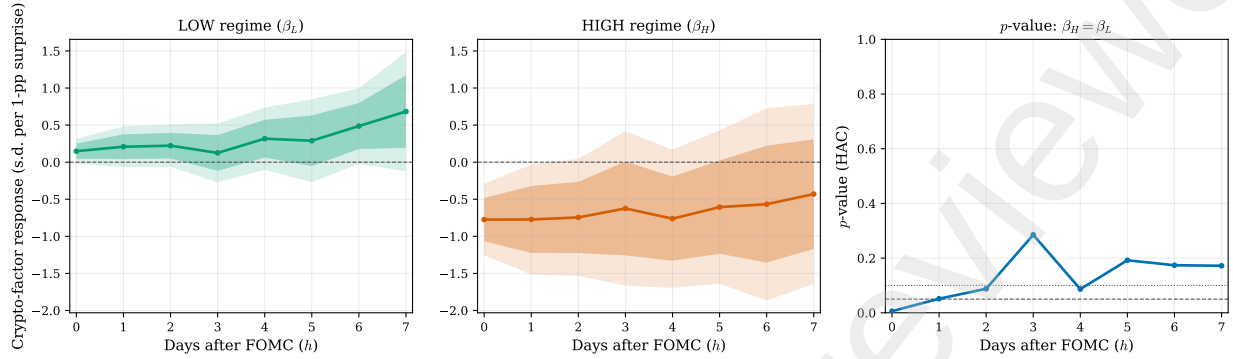
Figure B.9: LP-IV: Treasury-Yield Changes Instrumented by the High-Frequency Surprise



Notes: Local-projection instrumental-variable responses of the (standardized) global crypto factor to a one-percentage-point rise in the 1-year (top panel) and 2-year (bottom panel) Treasury yield, with the announcement-day yield change instrumented by the high-frequency monetary-policy surprise. Two-stage least squares on FOMC days with the baseline controls of equation (3); Newey–West (HAC) standard errors; 68% and 90% confidence bands; robust first-stage F statistics in the panel titles. Daily sample 2018–2026Q1 ($n = 71$ FOMC events); monetary-policy surprises from Acosta et al. (2025); Treasury yields from Eikon/Thomson Reuters.

The regime split of Section 4.2 does not hinge on restricting the coefficients on the controls to be common across regimes. Figure B.10 re-estimates the regime-interacted local projection in the fully state-interacted form of the state-dependent fiscal-multiplier literature (Ramey and Zubairy, 2018), interacting the intercept, the shock, and every control with the regime weights F_t and $1 - F_t$, so that the entire specification is allowed to differ across regimes. The pattern is unchanged—the low-regime response is indistinguishable from zero while the high-regime response is significantly negative—and the regime difference on impact is even sharper than in the baseline ($p = 0.006$).

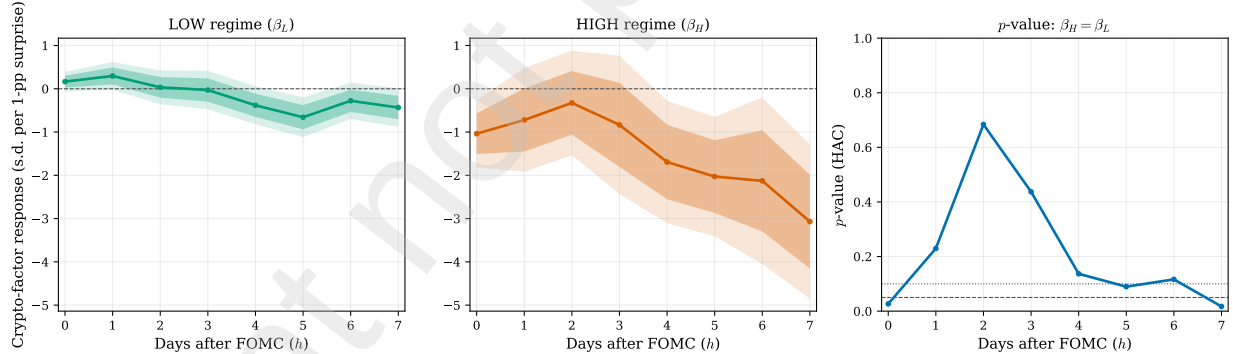
Figure B.10: Regime-Dependent Responses: Fully State-Interacted Specification



Notes: As in Figure 8, but with every regressor—the intercept, the shock, and each control—interacted with the regime weights F_t and $1 - F_t$, following Ramey and Zubairy (2018); the post-2020 block is subsumed by the regime-specific intercepts and slopes. HAC standard errors; 68% and 90% confidence bands; the right panel plots the HAC p -value of the regime difference (dashed line at 0.05, dotted at 0.10). Data, sample and sources as in Figure 8.

The split is also robust to computing the institutional-share proxy from exchange-related transactions (centralized and decentralized exchanges) only (Figure B.11).

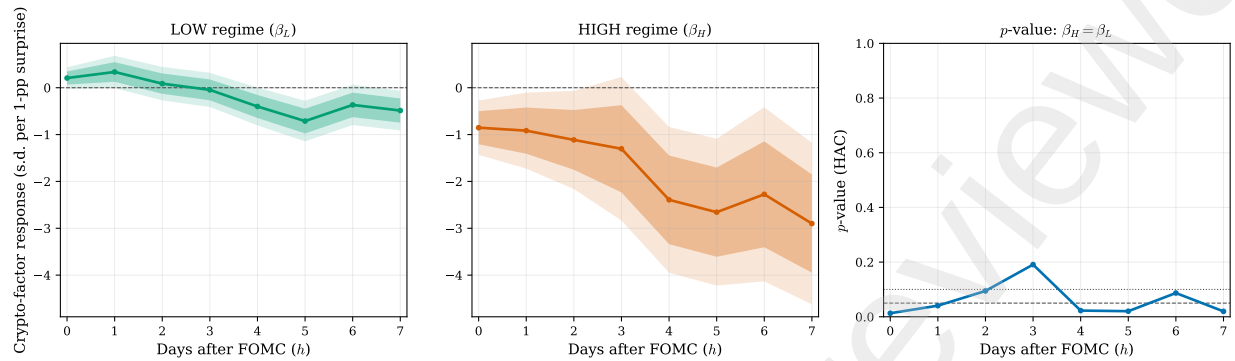
Figure B.11: Robustness: Exchange-Only Institutional Share



Notes: As in Figure 8, with the transition variable computed from the share of exchange-related transactions only. Monetary-policy surprises from Acosta et al. (2025); institutional share from Chainalysis.

The regime split is also robust to adding a smooth (cubic) time trend to the baseline controls (Figure B.12).

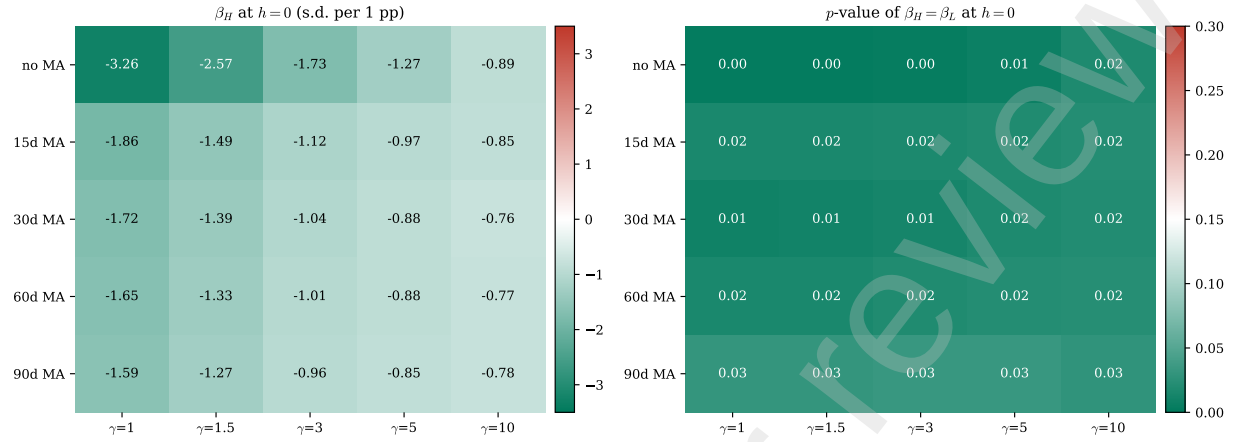
Figure B.12: Regime-Dependent Responses: Smooth Time Trend



Notes: As in Figure 8, with a smooth cubic time trend added to the baseline controls. Data, sample and sources as in Figure 8.

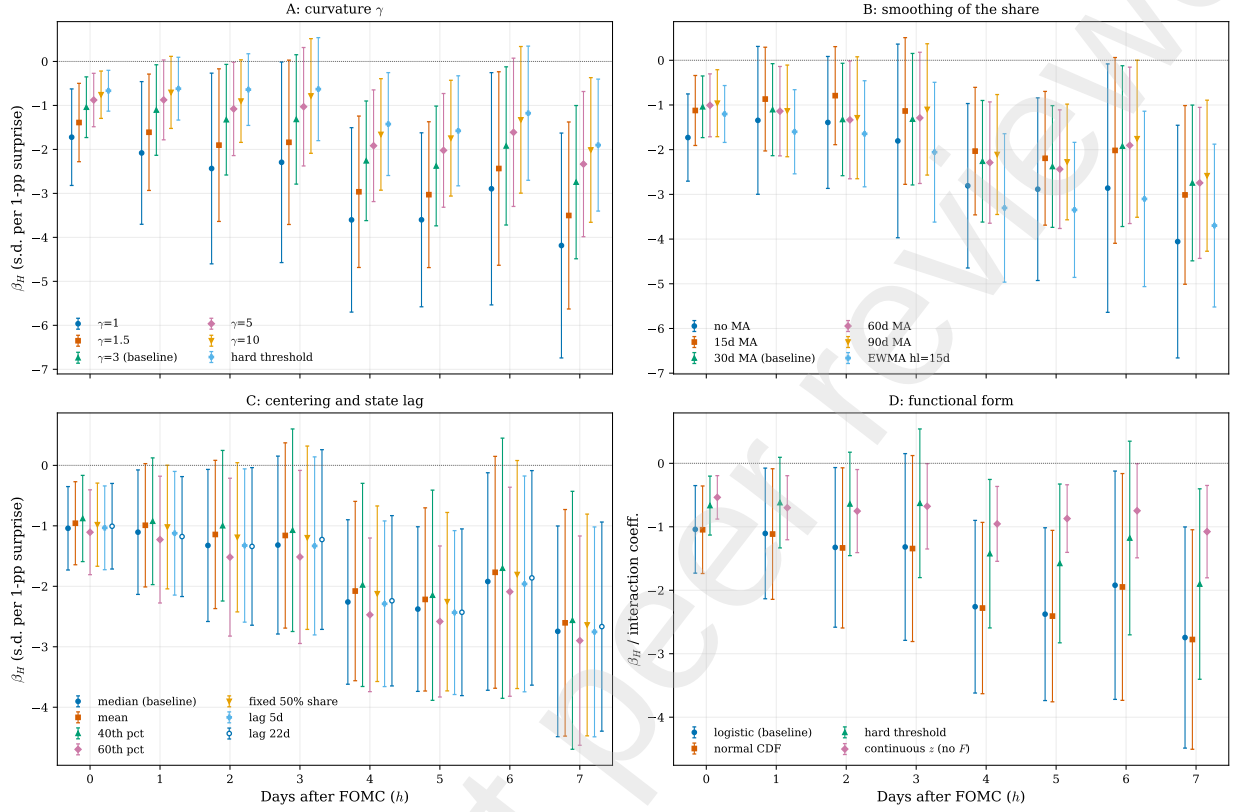
The split is also robust to the construction of the regime transition. Figure B.13 sweeps the two main parameters jointly—the smoothing of the institutional share (no smoothing to a 90-day moving average) and the curvature γ of the logistic transition (1 to 10)—and reports, for each of the 25 combinations, the high-regime impact response β_H and the p -value of the regime difference: β_H is negative in every cell and the regime difference is significant at the 5% level in every cell. Figure B.14 then varies every remaining ingredient of the transition one at a time—curvature (including a hard threshold), smoothing, centering and lag of the state, and the functional form (logistic, normal CDF, hard threshold, and a fully continuous interaction with no transition function)—with the same conclusion.

Figure B.13: Sensitivity of the Regime Split: Joint Parameter Sweep



Notes: Joint sweep of the regime-transition parameters (smoothing window $\times$ logistic curvature γ): high-regime impact response β_H (left panel) and p -value of the regime difference (right panel), at $h=0$. Specification otherwise as in Figure 8. Data and sample as in Figure 8.

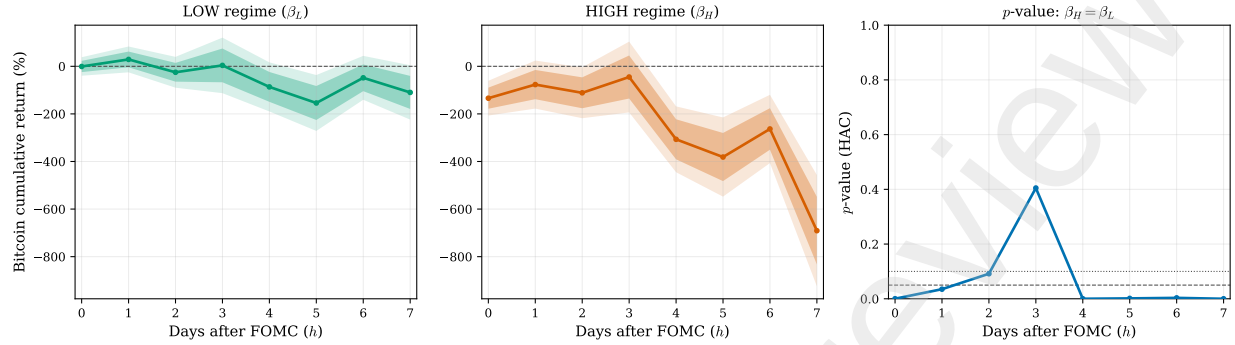
Figure B.14: Sensitivity of the Regime Split: One-at-a-Time Variations



Notes: One-at-a-time sensitivity of the high-regime response β_H to the regime-transition construction: curvature (A), smoothing (B), centering and state lag (C), and functional form (D; the continuous variant is per one standard deviation of the institutional share). Specification otherwise as in Figure 8; HAC standard errors; 90% confidence intervals. Data and sample as in Figure 8.

Figure B.15 re-estimates the regime-interacted local projection of Section 4.2 using the (log) Bitcoin price in place of the crypto factor. Since the Bitcoin price involves no factor estimation, this shows that the factor construction is not driving the results.

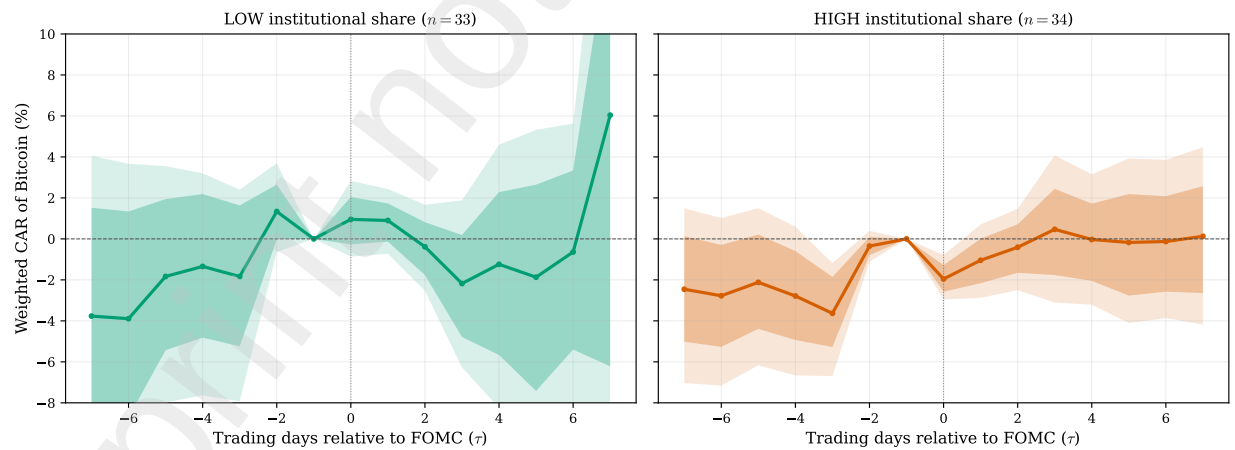
Figure B.15: Regime-Dependent Responses of the Bitcoin Price



Notes: Regime-interacted local-projection responses of the Bitcoin price ($100 \times \log$, so the coefficients read as cumulative percentage returns per one-percentage-point surprise) to a US monetary-policy surprise, by institutional-investor share: low regime (β_L), high regime (β_H), and the HAC p -value of their difference ($H_0: \beta_H = \beta_L$; dashed line at 0.05, dotted at 0.10). Bitcoin-only data; the specification (shock, transition, controls, sample) is as in Figure 8. Bitcoin prices from CryptoCompare; monetary-policy surprises from Acosta et al. (2025).

Finally, Figure B.16 replicates the event study of Section 4.2 using Bitcoin returns: the cumulative abnormal returns around FOMC announcements are flat in the low-institutional-participation regime, while in the high-participation regime Bitcoin drops significantly on impact.

Figure B.16: Event Study of Bitcoin around FOMC Announcements, by Institutional Regime



Notes: Sign-aligned, magnitude-weighted cumulative abnormal returns of Bitcoin (in percent) around FOMC announcements, as in equation (4), computed separately for meetings occurring in the high- and low-institutional-participation regimes. Shaded areas are 68% and 90% bootstrap confidence bands. FOMC meetings 2018–July 2025; Bitcoin prices from CryptoCompare; monetary-policy surprises from Acosta et al. (2025).